A Generalized Approach to Root-based Attacks against PLWE

Iván Blanco Chacón[1[0000−0002−4666−019X]], Raúl Durán Díaz[2[0000−0001−6217−4768]], and Rodrigo Martín Sánchez-Ledesma[3,4[0009−0001−1845−2959]]

[1] Departamento de Física y Matemáticas, Universidad de Alcalá, Spain
ivan.blancoc@uah.es
[2] Departamento de Automática, Universidad de Alcalá, Spain
raul.duran@uah.es
[3] Departamento de Álgebra, Universidad Complutense de Madrid, Spain
rodrma01@ucm.es
[4] Indra Sistemas de Comunicaciones Seguras, Spain
rmsanchezledesma@indra.es

Abstract. In the present work we address the robustness of the Polynomial Learning With Errors problem extending previous results in [3] and in [13]. In particular, we produce two kinds of new distinguishing attacks: a) we generalize [3] to the case where the defining polynomial has a root of degree up to 4, and b) we widen and refine the most general attack in [13] to the non-split case and determine further dangerous instances previously not detected. Finally, we exploit our results in order to show vulnerabilities of some cryptographically relevant polynomials.

Keywords: PLWE · Number Theory · Algebraic Roots · Trace-based Cryptanalysis.

1 Introduction

The Ring Learning With Errors problem (RLWE) and the Polynomial Learning With Errors problem (PLWE) sustain a large number of lattice-based cryptosystems highly believed to be quantum resistant. Some of the strongest theoretical clues pointing in this direction are the *worst case-average case* reductions from an approximate version of the Shortest Vector Problem on ideal lattices to the RLWE problem, established in [15], and to the PLWE problem over power-of-two cyclotomic polynomials, established in [20].

Furthermore, for a large family of monic irreducible polynomials $f(x) \in \mathbb{Z}[x]$, it is now well known that the PLWE problem for the ring $R_q := \mathbb{F}_q[x]/(f(x))$ is equivalent to the RLWE problem for the ring $\mathcal{O}_f/q\mathcal{O}_f$ where $\mathcal{O}_f$ is the ring of integers of the splitting field of $f(x)$ and q is a suitable rational prime. By *equivalent* it is understood that a solution to the first problem can be turned into a solution of the second (and vice versa) by an algorithm of polynomial complexity in the degree of $f(x)$, causing a noise increase which is also polynomial in the degree (see [19, Section 4]).

Cryptographically relevant families for which both problems admit simultaneously a *worst case-average case* reduction include the prime power cyclotomic case ([10], [11]) or, more in general, cyclotomic fields whose conductor is divisible by a small number of primes ([2]), as well as the maximal totally real cyclotomic subextensions of conductor $2^r pq$ for $r \geq 2$ and p, q primes or 1 ([4]).

However, despite the aforementioned worst case reductions, several sets of instances have been proved to be insecure in a number of works. For instance, in [13] and [14], a set of conditions has been identified granting an efficient attack against the search version of PLWE (hence against RLWE whenever they are equivalent). These conditions are of three types: *a*) the existence of an $\mathbb{F}_q$-root of $f(x)$ of *small multiplicative order*; *b*) the existence of an $\mathbb{F}_q$-root of *small canonical residue* in $\{0, \ldots, q-1\}$; and *c*) the existence of an $\mathbb{F}_q$-root such that the evaluation modulo q of the error polynomials over such root falls in the interval $\left[-\frac{q}{4}, \frac{q}{4}\right)$ with probability beyond $1/2$.

Further, in [8] the authors introduce the Chi-Square decisional attack on PLWE. This test detects non-uniformity of the samples modulo some of the prime ideals which divide q. For Galois number fields, this is enough to detect non-uniformity modulo q by virtue of the Chinese Remainder Theorem. Moreover, the authors show that their attack succeeds over some cyclotomic rings and subrings (although the ring of integers of the maximal totally real subextension is not listed therein). The attack requires typically beyond 10000 samples and may take a computation time up to thousands of hours.

Still, in [7], the attacks of [8], [13] and [14] are dramatically improved for the family of trinomials $x^n + ax + b$; indeed, the number of samples is reduced to 7 and the success rate is increased to 100%. Moreover, the authors justify that the reason why the attack succeeds is that the error distribution of this family is very skewed in certain directions, so that a moderate-to-large distortion between the coordinate and canonical embedding grants the success of the attack for small-to-moderate error sizes.

Finally, [18] extends the above geometric justification to all the vulnerable families in [13] and shows why the occurrence of this phenomenon suffices for the attack to work. In short, it shows that a generalized statistical attack works if there exists an element of small enough norm in the dual of a prime ideal dividing q (see [18, Lemma 3]).

These attacks, however, do not apply to any of the three lattice-based NIST selections for standardization (ML-KEM [17], ML-DSA [16] and FN-DSA) as they rely either on particular instances of Module Learning With Errors resistant problem, or on NTRU lattices, both not the target of these attacks. Moreover, these attacks work for the non-dual version of RLWE and for small values of the Gaussian parameter of the error distribution, while the worst case reduction of [15] holds for the dual version, where a very precise lower bound for the Gaussian parameter is required.

A more worrying future potential threat to real-world cryptosystems, even if theoretical, is the line of research which addresses the hardness of the ideal lattice Shortest Vector Problem itself. In this spirit, the works [9] and [12] exploit the Stickelberger class relations on cyclotomic fields to reduce the complexity of this problem to $\tilde{\mathcal{O}}(n^{1/2})$ and provide an explicit computation of the asymptotic constant involved.

The present work started as an extension of [1] and [3], which exploit the existence of a root α of $f(x)$ over finite non-trivial $\mathbb{F}_q$-extensions with small order trace. These works are in turn generalizations of [8] and [14], which assume the existence of small order roots in $\mathbb{F}_q$.

In both [1] and [3], the key tool is to use the composition of the evaluation-at-α with the trace map, in order to project R_q to a subring $R_{q,0}$ of large enough dimension over which the PLWE problem can be solved in expected polynomial time with overwhelming probability of success. In [1], the root is assumed to be quadratic and the attack on $R_{q,0}$-samples is pulled back to R_q, allowing for an attack on the usual PLWE, also with overwhelming probability of success in expected polynomial time. In [3], an attack on $R_{q,0}$-samples is obtained for cyclotomic polynomials of prime-power conductor but it remains still unclear how this attack can be pulled back to the original ring.

Hence, our original motivation was to extend these attacks to the case where there exists a root of higher order and where the attack can be pulled back to the whole ring. After succeeding in this task, we soon also observed that our trace-based approach allows to extend all the attacks of [8] and [14] (not just those based on the root orders) and, moreover, that it is still possible to refine one of the original algorithms and extend it to the higher-order root case in order to identify additional vulnerable instances of the PLWE problem.

Our work is organized as follows:

After some preliminaries in Section 2, Section 3 reintroduces the framework of [13], considering both truncation and no truncation on the range of the Gaussian distributions to appear in the PLWE problems. We believe this additional consideration helps understand the true nature of the attacks and might prove to be helpful when applied to real PLWE instances.

This section also introduces a revision of certain probability values assumed on [13,14] thus providing not only a better understanding of the applicability of the attacks, but also revealing further instances that yield to such attacks.

In Section 4 we extend the trace-based attack given in [1] for the quadratic root case to the higher-order root case (Proposition 12), dealing with the case when the root is of small order, thus giving rise to the *small set attack*.

In Proposition 14 we prove that the expected time of our algorithm is polynomial on the order of the root. We have tested our algorithm with some explicit instances obtaining, for one of them, a success probability of 0.629 with 350 test samples, and up to 0.993 with 500 samples (see Subsection 4.3, instance (2)).

Finally, in Section 5 we apply the ideas of Section 4 to the case when the evaluation of the error polynomials at the root gives values within the interval $\left[-\frac{q}{4}, \frac{q}{4}\right)$, which we term the *small values attack*.

2 Preliminaries

Let q be a prime and $f(x) \in \mathbb{Z}[x]$ a monic irreducible polynomial over $\mathbb{Z}[x]$ of degree N. Denote by R_q the ring $\mathbb{F}_q[x]/(f(x))$. For the PLWE distribution, we assume a Gaussian distribution of mean 0 and a certain variance σ^2.

We introduce here the following auxiliary lemmas, that will be useful throughout this work:

Lemma 1. *Given a discrete uniform random variable U in $\mathbb{F}_q$, where $q > 2$ is prime, and a discrete Gaussian random variable E of mean 0 and variance σ^2 in $\mathbb{F}_q$, we have that* $\mathrm{P}(U + E \mod q \in [-\frac{q}{4}, \frac{q}{4})) = \frac{1}{2} \pm \frac{1}{2q}$.

Proof. Expanding over the possible values of $e \mod q$, we have

$$\mathrm{P}(U + E \mod q \in [-\frac{q}{4}, \frac{q}{4})) =$$
$$\sum_{k=-\frac{q}{2}}^{\frac{q}{2}} \mathrm{P}(U + E \mod q \in [-\frac{q}{4}, \frac{q}{4}) \,|\, E \mod q = k) \cdot \mathrm{P}(E \mod q = k) =$$
$$\sum_{k=-\frac{q}{2}}^{\frac{q}{2}} \mathrm{P}(U \mod q \in [-\frac{q}{4} - k, \frac{q}{4} - k)) \cdot \mathrm{P}(E \mod q = k)$$

Now, it is required to analyze, given $k \in [-\frac{q}{2}, \frac{q}{2})$, $\mathrm{P}(U \mod q \in [-\frac{q}{4} - k, \frac{q}{4} - k))$. This analysis must be done in terms of the value of q, specifically regarding its congruence modulo 4:

If $q \equiv 1 \bmod 4$, then $\exists p$ such that $q = 4p + 1$:

$$\mathrm{P}(U \mod q \in [-\frac{q}{4} - k, \frac{q}{4} - k)) = \mathrm{P}(U \mod q \in [-p - k - \frac{1}{4}, p - k + \frac{1}{4}))$$

which means that U can take any value in the (discrete) interval $[-p-k, p-k]$, which represents $2p+1$ values in $\mathbb{F}_q$. That means that the probability is equal to

$$\frac{2p+1}{q} = \frac{\frac{q-1}{2} + 1}{q} = \frac{1}{2} + \frac{1}{2q}$$

On the other hand, if $q \equiv 3 \bmod 4$, then $\exists p$ such that $q = 4p + 3$ and the same argument yields that the probability is equal to

$$\frac{2p+1}{q} = \frac{\frac{q-3}{2} + 1}{q} = \frac{1}{2} - \frac{1}{2q}$$

Therefore, we have

$$\sum_{k=\frac{q}{2}}^{\frac{q}{2}} \mathrm{P}(U \mod q \in [-\frac{q}{4} - k, \frac{q}{4} - k)) \cdot \mathrm{P}(E \mod q = k) =$$
$$(\frac{1}{2} \pm \frac{1}{2q}) \cdot \sum_{k=-\frac{q}{2}}^{\frac{q}{2}} \mathrm{P}(E \mod q = k) = \frac{1}{2} \pm \frac{1}{2q}$$

as the sum over the possible values of E, taken modulo q, will be in the discrete interval $[-\frac{q}{2}, \frac{q}{2})$ and therefore that sum is equal to 1. ◻

The proof of the above lemma yields the following corollary, which will also be employed throughout the work:

Corollary 1. *Given a discrete uniform random variable U in $\mathbb{F}_q$, where $q > 2$ is prime, we have that* $\mathrm{P}(U \mod q \in [-\frac{q}{4}, \frac{q}{4})) = \frac{1}{2} \pm \frac{1}{2q}$.

Lemma 2. *If $X_1, X_2, \ldots, X_n$ are independent uniform distributions over $\mathbb{F}_{q^k}$ then, for each λ_1, λ_2, $\ldots$, $\lambda_n \in \mathbb{F}_{q^k}$ not all of them zero, the variable $\sum_{i=1}^{n} \lambda_i \cdot X_i$ is uniform*

Proof. The fact that the random variable $\lambda \cdot X$ is uniform in $\mathbb{F}_{q^k}$, if $\lambda \neq 0$ and X is uniform in $\mathbb{F}_{q^k}$ is a direct consequence of working in a field.

For X_1, X_2 independent uniforms in $\mathbb{F}_{q^k}$, we have

$$P[X_1 + X_2 = c] = \sum_{x_2 \in \mathbb{F}_{q^k}} P[X_1 = c - x_2 \wedge X_2 = x_2] = \frac{1}{q^k} \sum_{x_2 \in \mathbb{F}_{q^k}} P[X_2 = x_2] = \frac{1}{q^k}$$

When put together, we arrive to the desired result. □

3 Original $\mathbb{F}_q$-based root attacks, Revisited

In this section, we lay out the general setting that we will employ throughout our work, in the hope that it will be both more intuitively and more closely related to real-life deployments of PLWE-based schemes. This setting assumes that q is a prime and $f(x) \in \mathbb{Z}[x]$ a monic irreducible polynomial over $\mathbb{Z}[x]$ of degree N. Denote by R_q the ring $\mathbb{F}_q[x]/(f(x))$.

We will treat separately the cases where the PLWE Gaussian distribution of mean 0 and variance σ^2 is either assumed to be truncated to width 2σ or not truncated, contrary to [13,14], which only consider the truncated case. The rationale for this modification is that in real-life implementations of PLWE-based schemes the distributions are often not truncated. Moreover, we will see that this modification will impact the success probability of the attacks.

To this end, let $\mathcal{G}_\sigma$ be a centered Gaussian random variable $\mathcal{G}_\sigma$ of standard deviation σ truncated at 2σ or not. Let p_0 denote, from now on, the probability that $\mathcal{G}_\sigma$ takes values on the interval $[-2\sigma, 2\sigma]$. If $\mathcal{G}_\sigma$ is truncated to 2σ then $p_0 = 1$ and otherwise $p_0 \approx 0.954500$ (with error less than 10^{-6}).

3.1 Small Set Attack, Revisited

The authors of [13] give an attack on the decisional version of PLWE for R_q whenever a root $\alpha \in \mathbb{F}_q$ of $f(x)$ with (small) order r in $\mathbb{F}_q^*$ exists such that the cardinal of the set of all error polynomials evaluated at α, which we denote by Σ, is small enough for the set to be swept through.

Algorithm 1, as proposed in [13], is given a collection S of M samples, which come either from a uniform distribution, U, or from a PLWE distribution, $\mathcal{G}_\sigma$, assuming that $\mathrm{P}(D = \mathcal{G}_\sigma) = \mathrm{P}(D = U) = 1/2$, i.e., both distributions are equiprobable. The Algorithm decides which of both distributions the samples in collection S were drawn from, either from PLWE (returning a guess, g, for the evaluation of the secret $s(\alpha)$), or from the uniform one. Sometimes the algorithm is not able to decide, then returning **NOT ENOUGH SAMPLES**.

Input:	A collection of samples $S = \{(a_i(x), b_i(x))\}_{i=1}^{M} \subseteq R_q^2$
	A look-up table Σ of all possible values for $e(\alpha)$
Output:	A guess $g \in \mathbb{F}_q$ for $s(\alpha)$,
	or **NOT PLWE**,
	or **NOT ENOUGH SAMPLES**

- $G := \emptyset$
- ***for*** $g \in \mathbb{F}_q$ ***do***
  - ***for*** $(a_i(x), b_i(x)) \in S$ ***do***
    - ***if*** $b_i(\alpha) - a_i(\alpha)g \notin \Sigma$ ***then next*** g
  - $G := G \cup \{g\}$
- ***if*** $G = \emptyset$ ***then return*** **NOT PLWE**
- ***if*** $G = \{g\}$ ***then return*** g
- ***if*** $|G| > 1$ ***then return*** **NOT ENOUGH SAMPLES**

Algorithm 1. Algorithm for Small Set Attack over $\mathbb{F}_q$

The reason for the algorithm's applicability is that if the order r of the root α is low and the discrete Gaussian distribution is truncated, then the cardinal of the set Σ of plausible values for $e(\alpha)$

as $e(x)$ runs over the error distribution is small enough with respect to $|\mathbb{F}_q|$. Remark that

$$e(\alpha) = \sum_{j=0}^{r-1} e_j^* \alpha^j \text{ where } e_j^* = \sum_{i=0}^{\lfloor \frac{N}{r} \rfloor - 1} e_{ir+j},$$

where $\lfloor \ \rfloor$ stands for the floor function.

Notice that each term e_j^* follows a Gaussian distribution of mean 0 and variance $\overline{\sigma}^2 = \lfloor \frac{N}{r} \rfloor \cdot \sigma^2$. Therefore, we can ensure that for each $j \in \{0, ..., r-1\}$, we have that $|e_j^*| \leq 2\overline{\sigma}$ with probability at least p_0.

As explained above, let us define

$$\Sigma := \left\{ \sum_{j=0}^{r-1} x_j \alpha^j : \ x_j \in [-2\overline{\sigma}, 2\overline{\sigma}] \cap \mathbb{Z} \right\},$$

namely, the set of all possibilities for $e(\alpha)$, such that the absolute value of each block e_j^* is bounded by $2\overline{\sigma}$.

To begin with, we will consider the case where Gaussian distributions are truncated to width 2σ, where σ is their respective standard deviation. Then, we have the following

Proposition 1. *Let Σ, M and r be defined as above, and assume $|\Sigma| < q$. Then*

1. *If Algorithm 1 returns **PLWE** or **NOT ENOUGH SAMPLES**, then the samples are **PLWE** with probability at least $1 - q\left(\frac{|\Sigma|}{q}\right)^M$.*
2. *If Algorithm 1 returns **NOT PLWE**, then the samples are uniform with probability 1.*

Proof. Let us prove both items in turn.

1. Denote with E the event "Algorithm 1 returns **PLWE** or **NOT ENOUGH SAMPLES**". Then, applying Bayes' theorem,

$$\begin{aligned} \mathrm{P}(D = \mathcal{G}_\sigma | E) = 1 - \mathrm{P}(D = U|E) &= 1 - \frac{\mathrm{P}(E|D = U) \cdot \mathrm{P}(D = U)}{\mathrm{P}(E|D = U) \cdot \mathrm{P}(D = U) + \mathrm{P}(E|D = \mathcal{G}_\sigma) \cdot \mathrm{P}(D = \mathcal{G}_\sigma)} \\ &= 1 - \frac{\mathrm{P}(E|D = U)}{\mathrm{P}(E|D = U) + \mathrm{P}(E|D = \mathcal{G}_\sigma)}. \end{aligned} \tag{1}$$

Now remark that $\mathrm{P}(E|D = \mathcal{G}_\sigma)$ is the probability that at least a $g \in \mathbb{F}_q$ exists such that $b_i(\alpha) - a_i(\alpha)g \in \Sigma$, for all $i = 1, \ldots, M$, i.e., for all samples in S. Since we are supposing that the distribution is $\mathcal{G}_\sigma$ and Gaussian distributions to be truncated by 2σ, at least one g is sure to exist, namely, $s(\alpha)$, the evaluation of the secret. Then we have that $\mathrm{P}(E|D = \mathcal{G}_\sigma) = 1$.
As for $\mathrm{P}(E|D = U)$, let U_g be the event that $b_i(\alpha) - a_i(\alpha)g \in \Sigma$, for all $i = 1, \ldots, M$, i.e., for all samples in S. Then

$$\mathrm{P}(E|D = U) = \mathrm{P}\left(\bigcup_{g=0}^{q-1} U_g \right) \leq \sum_{g=0}^{q-1} \mathrm{P}(U_g),$$

where the inequality comes from the fact that we cannot guarantee that the events U_g be independent for different values of g.
Since we are now in the case where a_i and b_i are uniformly randomly selected, so $b_i - a_i g$ is also uniformly random according to Lemma 2; this means that $\mathrm{P}(b_i - a_i g \in \Sigma) = |\Sigma|/q$ for any i and any g, so

$$\mathrm{P}(U_g) = \prod_{i=1}^{M} \frac{|\Sigma|}{q} = \left(\frac{|\Sigma|}{q}\right)^M,$$

and thus we have

$$\mathrm{P}(E|D = U) \leq q\left(\frac{|\Sigma|}{q}\right)^M.$$

Plugging all the results above into Equation (1), we eventually obtain

$$\mathrm{P}(D = \mathcal{G}_\sigma|E) = 1 - \frac{\mathrm{P}(E|D = U)}{\mathrm{P}(E|D = U) + 1} \geq 1 - \mathrm{P}(E|D = U) \geq 1 - q\left(\frac{|\Sigma|}{q}\right)^M.$$

2. Denote now with E' the event "Algorithm 1 returns **NOT PLWE**". Applying again Bayes' theorem,

$$\begin{aligned}\mathrm{P}(D=U|E') &= \frac{\mathrm{P}(E'|D=U)\cdot\mathrm{P}(D=U)}{\mathrm{P}(E'|D=U)\cdot\mathrm{P}(D=U)+\mathrm{P}(E'|D=\mathcal{G}_\sigma)\cdot\mathrm{P}(D=\mathcal{G}_\sigma)}\\ &= \frac{\mathrm{P}(E'|D=U)}{\mathrm{P}(E'|D=U)+\mathrm{P}(E'|D=\mathcal{G}_\sigma)}.\end{aligned}$$

Now in order to compute $\mathrm{P}(E'|D=\mathcal{G}_\sigma)$ remark that the event E' is complementary to the event E, defined in the previous item. Hence

$$\mathrm{P}(E'|D=\mathcal{G}_\sigma)=1-\mathrm{P}(E|D=\mathcal{G}_\sigma)=1-1=0.$$

Consequently,

$$\mathrm{P}(D=U|E')=1,$$

as required.

□

We deal now with the case where the Gaussian distribution is not truncated. Then we have the following

Proposition 2. *Let Σ, M and r be defined as above, and assume $|\Sigma|<qp_0^r$. Then*

1. *If Algorithm 1 returns **PLWE** or **NOT ENOUGH SAMPLES**, then the samples are **PLWE** with probability at least $1-q\left(\frac{|\Sigma|}{qp_0^r}\right)^M$.*
2. *If Algorithm 1 returns **NOT PLWE**, then the samples are uniform with probability at least $1/2$ for large enough M.*

Proof. Let us prove both items in turn.

1. As before, denote with E the event "Algorithm 1 returns **PLWE** or **NOT ENOUGH SAMPLES**". Applying again Bayes' theorem,

$$\mathrm{P}(D=\mathcal{G}_\sigma|E)=1-\mathrm{P}(D=U|E)=1-\frac{\mathrm{P}(E|D=U)}{\mathrm{P}(E|D=U)+\mathrm{P}(E|D=\mathcal{G}_\sigma)}\geq 1-\frac{\mathrm{P}(E|D=U)}{\mathrm{P}(E|D=\mathcal{G}_\sigma)}.$$

Let U_g be again the event that $b_i(\alpha)-a_i(\alpha)g\in\Sigma$, for all $i=1,\ldots,M$, i.e., for all samples in S. Then

$$\mathrm{P}(E|D=\mathcal{G}_\sigma)=\mathrm{P}\left(\bigcup_{g=0}^{q-1}U_g\right)\geq\mathrm{P}(U_{g^*}),$$

where $g^*=s(\alpha)$. Observe that $b_i(\alpha)-a_i(\alpha)g^*=e_i(\alpha)$. Since we are now dealing with the not truncated case, we have

$$\mathrm{P}(U_{g^*})=\prod_{i=1}^{M}\mathrm{P}(e_i(\alpha)\in\Sigma)\geq p_0^{Mr}.$$

Moreover, $\mathrm{P}(E|D=U)$ is the same as in the truncated case, since the uniform distribution is not affected by the truncation. Combining results,

$$\mathrm{P}(D=\mathcal{G}_\sigma|E)\geq 1-\frac{\mathrm{P}(E|D=U)}{\mathrm{P}(E|D=\mathcal{G}_\sigma)}\geq 1-\frac{\mathrm{P}(E|D=U)}{p_0^{Mr}}\geq 1-\frac{q\left(\frac{|\Sigma|}{q}\right)^M}{p_0^{Mr}}=1-q\left(\frac{|\Sigma|}{qp_0^r}\right)^M.$$

2. Denote again with E' the event "Algorithm 1 returns **NOT PLWE**". Applying Bayes' theorem,

$$\mathrm{P}(D=U|E')=\frac{\mathrm{P}(E'|D=U)}{\mathrm{P}(E'|D=U)+\mathrm{P}(E'|D=\mathcal{G}_\sigma)}.$$

Remark as before that the event E' is complementary to the event E, defined in the previous item. Hence

$$\mathrm{P}(E'|D=\mathcal{G}_\sigma)=1-\mathrm{P}(E|D=\mathcal{G}_\sigma)\leq 1-p_0^{Mr}.$$

Consequently,

$$\mathrm{P}(D=U|E') \geq \frac{\mathrm{P}(E'|D=U)}{\mathrm{P}(E'|D=U)+1-p_0^{Mr}},$$

which is greater than $1/2$ if and only if $\mathrm{P}(E'|D=U) \geq 1-p_0^{rM}$. Since

$$\mathrm{P}(E'|D=U) \geq 1-q\left(\frac{|\Sigma|}{q}\right)^M,$$

for the conclusion to hold it is enough to take M such that $q^{\frac{1}{M}}\frac{|\Sigma|}{q} \leq p_0^r$.

□

Success probabilities. Observe that from the discussion above, we can derive the success probability of Algorithm 1. For the truncated case, we have the following

Proposition 3. *With the same notations and assumptions as in Proposition 1,*

1. *If the samples are PLWE, Algorithm 1 guesses correctly with probability* 1.
2. *If the samples are uniform, Algorithm 1 guesses correctly with probability at least* $1-q\left(\frac{|\Sigma|}{q}\right)^M$.

Proof. The first item is equivalent to computing $\mathrm{P}(E|D=\mathcal{G}_\sigma)$, which is 1, according to the proof in Proposition 1. Analogously, the second item corresponds to computing $\mathrm{P}(E'|D=U)$, which in the same place is shown to be greater or equal to $1-q\left(\frac{|\Sigma|}{q}\right)^M$. □

Now for the not truncated case, we have the following

Proposition 4. *With the same notations and assumptions as in Proposition 2,*

1. *If the samples are PLWE, Algorithm 1 guesses correctly with probability at least* p_0^{Mr}.
2. *If the samples are uniform, Algorithm 1 guesses correctly with probability at least* $1-q\left(\frac{|\Sigma|}{q}\right)^M$.

Proof. The first item is equivalent to computing $\mathrm{P}(E|D=\mathcal{G}_\sigma)$, which now is at least p_0^{Mr}, according to the proof in Proposition 2. The second item is not affected by the truncation so the bound is the same as for the truncated case, namely, greater or equal to $1-q\left(\frac{|\Sigma|}{q}\right)^M$. □

An improvement for a higher number of samples. Remark that in the attack described above, one of the success probabilities actually decreases with the number of samples. Although shocking at first, this is a natural consequence of the non-truncated behavior, as with each sample we are asking for another tentative error value to maintain the truncated behavior, which is less likely to happen.

While the attack itself could be negatively influenced by a higher number of samples (in case the samples were originally PLWE distributed), this does not mean that additional information (i.e. more samples) cannot be employed to assist in the decision.

One could always find an acceptable value of $M := M_0$ (such that both success probabilities remain as high as possible) and run the attack multiple times for different combinations of M_0 samples, and make a decision based upon some established rule. This is precisely what the remainder of this section will provide.

The idea is to take Algorithm 1 as a sub-process of our final Decision Algorithm. Given M samples, an appropriate number of samples $M_0 < M$ will be chosen to be used within the sub-process. Then, it will be run $c := \lfloor M/M_0 \rfloor$ times, collecting the output results from each round. Informally, if Algorithm 1 outputs anything different from **NOT PLWE** a number of times greater than expected, the Decision Algorithm concludes that the samples were indeed PLWE. Otherwise, the Decision Algorithm outputs that the samples were drawn from the uniform distribution.

Thus, our first task is to decide on an acceptable threshold for the number of expected executions of Algorithm 1 as a sub-process in order to output **PLWE**. In Proposition 4, the success probability of guessing the PLWE distribution of the samples is characterized by:

$$\mathrm{P}_{\mathcal{G}_\sigma}(M_0) := \mathrm{P}(E|D=\mathcal{G}_\sigma) \geq p_0^{M_0 r}.$$

Therefore, given M samples, the sub-process must be executed c times, thus providing an expected number of outputs, T, different from **NOT PLWE** given by

$$T := c \cdot \mathrm{P}_{\mathcal{G}_\sigma}(M_0) \geq c \cdot p_0^{M_0 r}$$

in case the samples were actually PLWE.

With these values, we define the *Extended Small Set Attack* in Algorithm 2.

Input:	A collection of samples $S = \{(a_i(x), b_i(x))\}_{i=1}^{M} \in R_q^2$
	A choice M_0 for the number of samples of the sub-process
Output:	A guess for the distribution of the samples, either
	PLWE or **UNIFORM**

- $T := \lceil c \cdot p_0^{M_0 r} \rceil$
- $C := 0$
- ***for*** $j \in \{0, ..., c-1\}$ ***do***
  - $result :=$ SmallSetAttack $\left(S_j := \{(a_i(x), b_i(x))\}_{i=j\cdot M_0+1}^{(j+1)\cdot M_0}\right)$
    - ***if*** $result \neq$ **NOT PLWE** ***then***
      - $C := C + 1$
- ***if*** $C < T$ ***then return*** **UNIFORM**
- ***else return*** **PLWE**

Algorithm 2. Algorithm for Extended Small Set Attack

Under this attack, we have the following success prediction probabilities:

- Case $D = \mathcal{G}_\sigma$: It is required to analyze the probability $\mathrm{P}(C \geq T | D = \mathcal{G}_\sigma)$. By definition, this can be seen as $1 - \mathrm{P}(C < T | D = \mathcal{G}_\sigma)$, which in turn can be interpreted in terms of a cumulative binomial distribution. Therefore, we have that

$$\mathrm{P}(C \geq T | D = \mathcal{G}_\sigma) := 1 - F(T-1, c, \mathrm{P}_{\mathcal{G}_\sigma}(M_0)) \geq 1 - F(T-1, c, p_0^{M_0 r})$$

  since the cumulative binomial distribution enjoys decreasing monotonicity on the *success probability* parameter.
- Case $D = U$: Now $\mathrm{P}(C < T | D = U)$ is yet again given in terms of the cumulative binomial distribution, i.e., the probability of a binomial distribution of parameters $n = c$ and $p = \mathrm{P}_U(M_0)$, to output any of the values $\{0, 1, 2, \dots, T-1\}$. Therefore, we have:

$$\mathrm{P}(C < T | D = U) := F(T-1, c, \mathrm{P}_U(M_0)) \geq F(T-1, c, 1 - (|\Sigma|/q)^{M_0})$$

  since the probability of success in case the samples were uniform (i.e. $\mathrm{P}_U(M_0)$), in terms of the proof of Proposition 4, verifies

$$\mathrm{P}_U(M_0) = 1 - \mathrm{P}\left(\bigcup_{g=0}^{q-1} U_g\right) \geq 1 - \mathrm{P}\left(U_g\right) = 1 - (|\Sigma|/q)^{M}, \forall g \in \mathbb{F}_q$$

  and therefore can be upper-bounded by the probability that a particular $g_0 \in \mathbb{F}_q$ verifies the condition for $M = M_0$ samples, which is $1 - (|\Sigma|/q)^{M_0}$.

With the above characterization of the success probabilities of each distribution, we prove that Algorithm 2 is successful:

Proposition 5. *Let C, T, M_0 be as specified in Algorithm 2. Then, if $1 - (|\Sigma|/q)^{M_0} < p_0^{M_0 r}$, Algorithm 2 is successful.*

Proof. We need to prove that the algorithm verifies that:

$$\frac{1}{2}\cdot(\mathrm{P}(C<T|D=U)+\mathrm{P}(C\geq T|D=\mathcal{G}_\sigma))\geq$$
$$\frac{1}{2}+\frac{1}{2}\cdot(F(T-1,c,1-(|\Sigma|/q)^{M_0})-F(T-1,c,p_0^{M_0 r}))>\frac{1}{2}.$$

Employing the decreasing monotonicity of the cumulative binomial distribution, this is true as long as

$$1-(|\Sigma|/q)^{M_0}<p_0^{M_0 r}.$$

□

This result provides a characterization of how M_0 can be chosen in order for the attack to be applicable, although further refinements could be employed.

It is important to note that this extended algorithm does not make direct use of features of $\mathbb{F}_q$, other than those related to the algorithm being employed as sub-process. Therefore, this technique will also be applicable in the generalization to finite field extensions of Section 4.

3.2 Small Values Attack, Revisited

The authors of [13] give yet another attack on the decisional version of PLWE for R_q whenever a root $\alpha\in\mathbb{F}_q$ of $f(x)$ exists such that the evaluations of the error polynomials at α give "small" values.

Algorithm 3, as proposed in [13], is given a collection S of M samples, which come either from a uniform distribution, U, or from a PLWE distribution, $\mathcal{G}_\sigma$, assuming that $\mathrm{P}(D=\mathcal{G}_\sigma)=\mathrm{P}(D=U)=1/2$, i.e., both distributions are equiprobable. The Algorithm decides which one of both distributions the samples in collection S were drawn from, either from PLWE (returning a guess, g, for the evaluation of the secret $s(\alpha)$), or from the uniform one. Sometimes the algorithm is not able to decide, then returning **NOT ENOUGH SAMPLES**.

Input:	A collection of samples $S=\{(a_i(x),b_i(x))\}_{i=1}^M\in R_q^2$
Output:	A guess $g\in\mathbb{F}_q$ for $s(\alpha)$,
	or **NOT PLWE**,
	or **NOT ENOUGH SAMPLES**

- *set* $G:=\emptyset$
- *for* $g\in\mathbb{F}_q$ *do*
  - *for* $(a_i(x),b_i(x))\in S$ *do*
    - *if* $b_i(\alpha)-a_i(\alpha)g\notin[-\frac{q}{4},\frac{q}{4})$ *then* next g
  - *set* $G:=G\cup\{g\}$
- *if* $G=\emptyset$ *then return* **NOT PLWE**
- *if* $G=\{g\}$ *then return* g
- *if* $|G|>1$ *then return* **NOT ENOUGH SAMPLES**

Algorithm 3. Algorithm for Small Values Attack over $\mathbb{F}_q$

The reason of the algorithm's applicability is that the evaluation of the error polynomials at α yields new Gaussian distributions of mean 0 and a certain variance $\overline{\sigma}^2$. Moreover, the smaller the order of the root, the smaller the variance of the image Gaussian distribution.

Case $\boldsymbol{\alpha=\pm 1}$. In this case, the value $e_i(\alpha)=\sum_{j=0}^{n-1}e_{ij}\cdot\alpha^j$ is a sum of n Gaussians of mean 0 and variance σ^2, so it is a Gaussian of mean 0 and variance $n\cdot\sigma^2$. Therefore $\mathcal{G}_{\overline{\sigma}}$ is a Gaussian with $\overline{\sigma}^2=n\cdot\sigma^2$.

Case $\boldsymbol{\alpha\neq\pm 1}$ and has small order $\boldsymbol{r}$ modulo $\boldsymbol{q}$. Here, the small multiplicative order of the root α is used to group the error value $e(\alpha)$ into r packs of $\frac{n}{r}$ values. Each pack forms an independent Gaussian of mean 0 and variance $\frac{n}{r}\cdot\sigma^2$, and thus the overall distribution is a weighted sum of such Gaussians, which generates a Gaussian of mean 0 and variance $\overline{\sigma}^2=\sum_{i=0}^{r-1}\frac{n}{r}\cdot\sigma^2\cdot\alpha^{2\cdot i}$.

Case $\alpha \neq \pm 1$ and does not have small order modulo q. In the most general case, we just have a weighted sum of n Gaussians, which is itself a Gaussian of mean 0 and variance $\overline{\sigma}^2 = \sum_{i=0}^{n-1} \sigma^2 \cdot \alpha^{2 \cdot i}$.

Under this attack the key factor is that, if the bound that reigns over the Gaussian image does not surpass the value $\frac{q}{4}$, then the probability that for any sample i the evaluation of the error polynomial lies inside $[-\frac{q}{4}, \frac{q}{4})$ is 1 for $g = s(\alpha) \bmod q$, under the supposition that the resulting Gaussian distribution is truncated.

Below we will deal in turn with both truncated and not truncated Gaussian distributions. To begin with, we deal with the first one by stating the following

Proposition 6. *Assume $2\overline{\sigma} \leq \frac{q}{4}$ and let M be the number of input samples. Then*

1. *If Algorithm 3 returns **PLWE** or **NOT ENOUGH SAMPLES**, then the samples are **PLWE** with probability at least $1 - q\left(\frac{1}{2} \pm \frac{1}{2q}\right)^M$.*
2. *If Algorithm 3 returns **NOT PLWE**, then the samples are uniform with probability 1.*

Proof. We follow essentially the same lines as in the proof of Proposition 1, dealing with both items in turn.

1. Denote with E the event "Algorithm 3 returns **PLWE** or **NOT ENOUGH SAMPLES**". Then, applying Bayes' theorem,
$$\mathrm{P}(D = \mathcal{G}_\sigma | E) = 1 - \mathrm{P}(D = U|E) = 1 - \frac{\mathrm{P}(E|D = U)}{\mathrm{P}(E|D = U) + \mathrm{P}(E|D = \mathcal{G}_\sigma)}. \tag{2}$$
Now remark that $\mathrm{P}(E|D = \mathcal{G}_\sigma)$ is the probability that at least a $g \in \mathbb{F}_q$ exists such that $b_i(\alpha) - a_i(\alpha)g \in [-\frac{q}{4}, \frac{q}{4})$, for all $i = 1, \ldots, M$, i.e., for all samples in S. Since we are supposing that the distribution is $\mathcal{G}_\sigma$, at least one g is sure to exist, namely, $g = s(\alpha)$, the evaluation of the secret, as $e_i(\alpha) \in [-2\overline{\sigma}, 2\overline{\sigma}] \subseteq [-\frac{q}{4}, \frac{q}{4})$ for each sample i. Then we have that $\mathrm{P}(E|D = \mathcal{G}_\sigma) = 1$.
As for $\mathrm{P}(E|D = U)$, let U_g be the event that $b_i(\alpha) - a_i(\alpha)g \in [-\frac{q}{4}, \frac{q}{4})$, for all $i = 1, \ldots, M$, i.e., for all samples in S. Then
$$\mathrm{P}(E|D = U) = \mathrm{P}\left(\bigcup_{g=0}^{q-1} U_g\right) \leq \sum_{g=0}^{q-1} \mathrm{P}(U_g),$$
where the inequality comes from the fact that we cannot guarantee that the events U_g be independent for different values of g.
Since we are now in the case where a_i and b_i are uniformly randomly selected, so $b_i - a_i g$ is also uniformly random according to Lemma 2. By applying Proposition 1, this means that for any i and any g, $\mathrm{P}(b_i - a_i g \in [-\frac{q}{4}, \frac{q}{4})) = \frac{1}{2} \pm \frac{1}{2q}$, so
$$\mathrm{P}(U_g) = \prod_{i=1}^{M} \frac{1}{2} \pm \frac{1}{2q} = \left(\frac{1}{2} \pm \frac{1}{2q}\right)^M,$$
and thus we have
$$\mathrm{P}(E|D = U) \leq \sum_{g=0}^{q-1} \mathrm{P}(U_g) = q\left(\frac{1}{2} \pm \frac{1}{2q}\right)^M.$$
Plugging all the results above into Equation (2), we eventually obtain
$$\mathrm{P}(D = \mathcal{G}_\sigma | E) = 1 - \frac{\mathrm{P}(E|D = U)}{\mathrm{P}(E|D = U) + 1} \geq 1 - \mathrm{P}(E|D = U) \geq 1 - q\left(\frac{1}{2} \pm \frac{1}{2q}\right)^M.$$
2. The proof for this item is exactly the same as the one for item 2 in Proposition 1.

□

Now we turn our attention to the not truncated case. We have the following

Proposition 7. *Assume $2\overline{\sigma} \leq \frac{q}{4}$ and $\frac{1}{2} \pm \frac{1}{2q} < p_0$. Let M be the number of input samples. Then*

1. *If Algorithm 3 returns **PLWE** or **NOT ENOUGH SAMPLES**, then the samples are **PLWE** with probability at least* $1 - q\left(\frac{\frac{1}{2} \pm \frac{1}{2q}}{p_0}\right)^M$.
2. *If Algorithm 3 returns **NOT PLWE**, then the samples are uniform with probability at least* $1/2$ *for large enough* M*.*

Proof. We follow essentially the same lines as in the proof of Proposition 2, dealing with both items in turn.

1. As before, denote with E the event "Algorithm 3 returns **PLWE** or **NOT ENOUGH SAMPLES**". Applying again Bayes' theorem,

$$\mathrm{P}(D = \mathcal{G}_\sigma | E) = 1 - \mathrm{P}(D = U|E) = 1 - \frac{\mathrm{P}(E|D = U)}{\mathrm{P}(E|D = U) + \mathrm{P}(E|D = \mathcal{G}_\sigma)} \geq 1 - \frac{\mathrm{P}(E|D = U)}{\mathrm{P}(E|D = \mathcal{G}_\sigma)}.$$

Let U_g be the event that $b_i(\alpha) - a_i(\alpha)g \in [-\frac{q}{4}, \frac{q}{4})$, for all $i = 1, \ldots, M$, i.e., for all samples in S. Then

$$\mathrm{P}(E|D = \mathcal{G}_\sigma) = \mathrm{P}\left(\bigcup_{g=0}^{q-1} U_g\right) \geq \mathrm{P}(U_{g^*}),$$

where $g^* = s(\alpha)$. Observe that $b_i(\alpha) - a_i(\alpha)g^* = e_i(\alpha)$. Since we are now dealing with the not truncated case, we have

$$\mathrm{P}(U_{g^*}) = \prod_{i=1}^{M} \mathrm{P}(e_i(\alpha) \in [-\frac{q}{4}, \frac{q}{4})) \geq p_0^M.$$

Moreover, $\mathrm{P}(E|D = U)$ is the same as in the truncated case, since the uniform distribution is not affected by the truncation. Combining results,

$$\mathrm{P}(D = \mathcal{G}_\sigma|E) \geq 1 - \frac{\mathrm{P}(E|D = U)}{\mathrm{P}(E|D = \mathcal{G}_\sigma)} \geq 1 - \frac{\mathrm{P}(E|D = U)}{p_0^M} \geq 1 - \frac{q\left(\frac{1}{2} \pm \frac{1}{2q}\right)^M}{p_0^M}.$$

2. The proof for this item is exactly the same as the one for item 2 in Proposition 1.

□

Success probabilities. Remark that from the discussion above, we can derive the success probability of Algorithm 3. For the truncated case, we have the following

Proposition 8. *With the same notations and assumptions as in Proposition 6,*

1. *If the samples are PLWE, Algorithm 3 guesses correctly with probability* 1.
2. *If the samples are uniform, Algorithm 3 guesses correctly with probability at least* $1 - q\left(\frac{1}{2} \pm \frac{1}{2q}\right)^M$.

Proof. The first item is equivalent to computing $\mathrm{P}(E|D = \mathcal{G}_\sigma)$, which is 1, according to the proof in Proposition 6. Analogously, the second item corresponds to computing $\mathrm{P}(E'|D = U)$, which in the same place is shown to be greater or equal to $1 - q\left(\frac{1}{2} \pm \frac{1}{2q}\right)^M$. □

Now for the not truncated case, we have the following

Proposition 9. *With the same notations and assumptions as in Proposition 7,*

1. *If the samples are PLWE, Algorithm 3 guesses correctly with probability at least* p_0^M.
2. *If the samples are uniform, Algorithm 3 guesses correctly with probability at least* $1 - q\left(\frac{1}{2} \pm \frac{1}{2q}\right)^M$.

Proof. The first item is equivalent to computing $\mathrm{P}(E|D = \mathcal{G}_\sigma)$, which now is at least p_0^M, according to the proof in Proposition 7. The second item is not affected by the truncation so the bound is the same as for the truncated case, namely, greater or equal to $1 - q\left(\frac{1}{2} \pm \frac{1}{2q}\right)^M$. □

An improvement for a higher number of samples. The same considerations over the number of samples present apply to this attack. So, in the remainder of this section the same techniques will be applied.

Thus, it is first required to revisit an acceptable threshold for the number of expected executions of the *small values attack* sub-process to output **PLWE**. In Proposition 9, the success probability of guessing the PLWE distribution of the samples is characterized by:

$$\mathrm{P}_{\mathcal{G}_\sigma}(M_0) := \mathrm{P}(E|D = \mathcal{G}_\sigma) \geq p_0^{M_0}.$$

Therefore, given M samples, the sub-process must executed $c := \lfloor M/M_0 \rfloor$ times, thus providing an expected number of outputs, T, different from **NOT PLWE** given by

$$T := c \cdot \mathrm{P}_{\mathcal{G}_\sigma}(M_0) \geq c \cdot p_0^{M_0}$$

in case the samples were actually PLWE.

With these values, we define the *Extended Small Values Attack* in Algorithm 4.

Input:	A collection of samples $S = \{(a_i(x), b_i(x))\}_{i=1}^{M} \in R_q^2$
	A choice M_0 for the number of samples of the sub-process
Output:	A guess into the distribution of the samples, either
	PLWE or **UNIFORM**

- $T := \lceil c \cdot p_0^{M_0} \rceil$
- $C := 0$
- ***for*** $j \in \{0, ..., c-1\}$ ***do***
  - $result := SmallValuesAttack\left(S_j := \{(a_i(x), b_i(x))\}_{i=j\cdot M_0+1}^{(j+1)\cdot M_0}\right)$
    - ***if*** result $\neq$ **NOT PLWE** ***then***
      - $C := C + 1$
- ***if*** $C < T$ ***then return*** **UNIFORM**
- ***else return*** **PLWE**

Algorithm 4. Algorithm for Extended Small Values Attack

Under this attack, we have the following success prediction probabilities, which closely follow those in Subsection 3.1:

- Case $D = \mathcal{G}_\sigma$: It is required to analyze the probability $\mathrm{P}(C \geq T|D = \mathcal{G}_\sigma)$. By definition, this can be seen as $1 - \mathrm{P}(C < T|D = \mathcal{G}_\sigma)$, which in turn can be interpreted in terms of a cumulative binomial distribution. Therefore, we have that

$$\mathrm{P}(C \geq T|D = \mathcal{G}_\sigma) := 1 - F(T-1, c, \mathrm{P}_{\mathcal{G}_\sigma}(M_0)) \geq 1 - F(T-1, c, p_0^{M_0})$$

  since the cumulative binomial distribution enjoys decreasing monotonicity on the *success probability* parameter.
- Case $D = U$: Now $\mathrm{P}(C < T|D = U)$ is yet again given in terms of the cumulative binomial distribution, that is, the probability of a binomial distribution of parameters $n = c$ and $p = \mathrm{P}_U(M_0)$, to output any of the values $\{0, 1, 2, \ldots, T-1\}$. Therefore, we have:

$$\mathrm{P}(C < T|D = U) := F(T-1, c, \mathrm{P}_U(M_0)) \geq F\left(T-1, c, 1 - \left(\frac{1}{2} \pm \frac{1}{2q}\right)^{M_0}\right),$$

  since the probability of success in case the samples were uniform (i.e. $\mathrm{P}_U(M_0)$), in terms of the proof of Proposition 9, verifies

$$\mathrm{P}_U(M_0) = 1 - \mathrm{P}\left(\bigcup_{g=0}^{q-1} U_g\right) \geq 1 - \mathrm{P}\left(U_g\right) = 1 - \left(\frac{1}{2} \pm \frac{1}{2q}\right)^{M}, \forall g \in \mathbb{F}_q$$

  and therefore can be upper-bounded by the probability that a particular $g_0 \in \mathbb{F}_q$ verifies the condition for $M = M_0$ samples, which is $1 - \left(\frac{1}{2} \pm \frac{1}{2q}\right)^{M_0}$.

With the above characterization of the success probabilities of each distribution, we prove that Algorithm 4 is successful:

Proposition 10. *Let C, T, M_0 be as specified in Algorithm 4. Then, if $1-\left(\frac{1}{2} \pm \frac{1}{2q}\right)^{M_0} < p_0^{M_0}$, Algorithm 4 is successful.*

Proof. We need to prove that the algorithm verifies that:

$$\frac{1}{2} \cdot (\mathrm{P}(C < T | D = U) + \mathrm{P}(C \geq T | D = \mathcal{G}_\sigma)) \geq$$
$$\frac{1}{2} + \frac{1}{2} \cdot \left(F\left(T-1, c, 1-\left(\frac{1}{2} \pm \frac{1}{2q}\right)^{M_0}\right) - F(T-1, c, p_0^{M_0})\right) > \frac{1}{2}$$

Employing the decreasing monotonicity of the cumulative binomial distribution, this is true as long as

$$1-\left(\frac{1}{2} \pm \frac{1}{2q}\right)^{M_0} < p_0^{M_0}.$$

□

This result provides a characterization of how M_0 should be chosen for the attack to be applicable, although further refinements could be employed.

As in the case of the *small set attack*, it is again noticeable that this extended algorithm does not make direct use of features of $\mathbb{F}_q$, other than those related to the algorithm being employed as sub-process. Therefore, this technique will also be applicable to the extension to finite field extensions of Section 5 (with only having to revisit, if necessary, the success probabilities of the sub-processes employed).

3.3 Unbounded Small Values Attack, Revisited

We now turn our attention to the case where $2\overline{\sigma} \geq \frac{q}{4}$. The value computed by evaluating at α a tentative error polynomial, $b_i(\alpha) - a_i(\alpha)g$, for a sample i and a guess g for $s(\alpha)$, no longer provides an automatic distinguisher, since this value will lie outside the interval $[-\frac{q}{4}, \frac{q}{4})$ with significant probability.

However, [14] shows an attack to cover the previous scenario, though considering a truncated Gaussian for the error polynomials. This section provides a detailed overview of the implications of lifting the latter requirement.

To construct this attack, the probability of the event E_i, defined as the event in which $b_i(\alpha) - a_i(\alpha)g \in [-\frac{q}{4}, \frac{q}{4})$, for the i-th sample, and a guess g for $s(\alpha)$, needs to be re-calculated, and we will do it by means of a plain "favorable vs total-cases" analysis.

Since we work modulo q, we have (discrete) values inside the interval $[0, q)$, which can equivalently be seen inside the interval $[-\frac{q}{2}, \frac{q}{2})$, by means of associating each value in the interval $[\frac{q}{2}, q)$ to its unique congruent value in $[-\frac{q}{2}, 0)$.

With this transformation, its easy to see that for any $b = a \bmod q$, b lies inside $[-\frac{q}{4}, \frac{q}{4})$ when

$$a \in \left(\bigcup_{j\in\mathbb{N}} [-\frac{5\cdot q}{4} - j\cdot q, -\frac{3\cdot q}{4} - j\cdot q)\right) \cup [-\frac{q}{4}, \frac{q}{4}) \cup \left(\bigcup_{j\in\mathbb{N}} [\frac{3\cdot q}{4} + j\cdot q, \frac{5\cdot q}{4} + j\cdot q)\right).$$

Hence, we can characterize the favorable cases by the area of the image distribution across the favorable space, namely,

$$\int_{-\frac{q}{4}}^{\frac{q}{4}} \mathcal{G}_{\overline{\sigma}} + \sum_{j\in\mathbb{N}} \int_{-\frac{5\cdot q}{4}-j\cdot q}^{-\frac{3\cdot q}{4}-j\cdot q} \mathcal{G}_{\overline{\sigma}} + \sum_{j\in\mathbb{N}} \int_{\frac{3\cdot q}{4}+j\cdot q}^{\frac{5\cdot q}{4}+j\cdot q} \mathcal{G}_{\overline{\sigma}}.$$

Now, the area of this distribution is symmetric across 0, so both integrals can be truncated to its positive side, eventually yielding the following probability:

$$\mathrm{P}(E_i | D = \mathcal{G}_\sigma) = \left(\int_0^\infty \mathcal{G}_{\overline{\sigma}}\right)^{-1} \cdot \left(\int_0^{\frac{q}{4}} \mathcal{G}_{\overline{\sigma}} + \sum_{j\in\mathbb{N}} \int_{\frac{3\cdot q}{4}+j\cdot q}^{\frac{5\cdot q}{4}+j\cdot q} \mathcal{G}_{\overline{\sigma}}\right).$$

Note that, the implication of the truncated behavior of the Gaussian distribution appears on the above probability: in a truncated setting, the integral sum has a finite span, which corresponds to the finite number of intervals inside $[-2\overline{\sigma}, 2\overline{\sigma}]$.

Denoting the value of this probability as $\mathrm{P}(E_i|D = \mathcal{G}_\sigma) = \frac{1}{2} + \delta$, and ℓ as the number of samples provided, the following attack applies:

Input:	A collection of samples $S = \{(a_i(x), b_i(x))\}_{i=1}^{\ell} \in R_q^2$
	A value δ
Output:	A guess into the distribution of the samples, either
	PLWE or **UNIFORM**

- $T := \left\lceil \frac{1}{2}\left(\ell \cdot q + 2\ell \cdot \delta \pm \ell \cdot \left(1 - \frac{1}{q}\right)\right)\right\rceil$
- $C := 0$
- *for* $g \in \mathbb{F}_q$ *do*
  - *for* $(a_i(x), b_i(x)) \in S$ *do*
    - *if* $b_i(\alpha) - a_i(\alpha)g \in [-\frac{q}{4}, \frac{q}{4})$ *then*
      - $C := C + 1$
- *if* $C < T$ *then return* **UNIFORM**
- *else return* **PLWE**

Algorithm 5. Algorithm for Unbounded Small Values Attack over $\mathbb{F}_q$

In Algorithm 5, T represents the number of times that, on average, one would expect the event E_i to hold true when the Algorithm is executed, should the samples actually come from the PLWE distribution.

To understand the value for this threshold T, consider that all candidates g from $\mathbb{F}_q$, except one, will not be the desired value $g^* = s(\alpha)$, and therefore the *tentative* error value

$$b(\alpha) - g \cdot a(\alpha) = (s(\alpha) - g) \cdot a(\alpha) + e(\alpha)$$

becomes the sum of a discrete uniform term in $\mathbb{F}_q$ and a discrete Gaussian element of mean 0 and variance $\overline{\sigma}^2$ over $\mathbb{F}_q$. Therefore applying Lemma 1, we have that the probability for that sum to lie inside $[-\frac{q}{4}, \frac{q}{4})$, for any given sample i and guess $g \neq g^*$, is $\frac{1}{2} \pm \frac{1}{2q}$.

Moreover, for one value, namely, the correct guess g^*, we have that the probability of the samples to lie inside the objective interval is $\frac{1}{2} + \delta$, for each sample i, by the hypothesis.

Therefore, the number of expected values inside the desired interval can be calculated as

$$T := \left\lceil \ell(q-1)\left(\frac{1}{2} \pm \frac{1}{2q}\right) + \ell\left(\frac{1}{2} + \delta\right)\right\rceil = \left\lceil \frac{1}{2}\left(\ell \cdot q + 2\ell \cdot \delta \pm \ell \cdot \left(1 - \frac{1}{q}\right)\right)\right\rceil.$$

Under this attack, we have the following success prediction probabilities:

- Case $D = U$: The probability $\mathrm{P}(C < T|D = U)$ is given by the cumulative binomial distribution, that is, the probability of a binomial distribution of parameters $n = \ell \cdot q$ and $p = \frac{1}{2} \pm \frac{1}{2q}$, to output any of the values $\{0, 1, 2, \ldots, T-1\}$. Therefore, we have:

$$\mathrm{P}(C < T|D = U) = F\left(T - 1, \ell \cdot q, \frac{1}{2} \pm \frac{1}{2q}\right).$$

- Case $D = \mathcal{G}_\sigma$: Now we need to analyze the probability $\mathrm{P}(C \geq T|D = \mathcal{G}_\sigma)$. Let C_{g^*} be defined as the number of samples inside the objective interval, given the correct guess g^*. Then:

$$\mathrm{P}(C \geq T|D = \mathcal{G}_\sigma) = \sum_{i=0}^{\ell} \mathrm{P}(C - C_{g^*} \geq T - i|D = \mathcal{G}_\sigma) \cdot \mathrm{P}(C_{g^*} = i). \tag{3}$$

  The first term of the sum is yet again the complementary probability of a cumulative binomial on all the guesses $g \neq g^*$, namely, those not the correct one. Consequently, they fall under the case of Lemma 1, and so its probability of satisfying the event is $\frac{1}{2} \pm \frac{1}{2q}$.

The second term is the probability that a binomial distribution, with $n = \ell$ and $p = \frac{1}{2} + \delta$ outputs the value i. Therefore, Equation (3) becomes

$$\mathrm{P}(C \geq T | D = \mathcal{G}_\sigma) = \sum_{i=0}^{\ell} \left(1 - F\left(T - i - 1, \ell \cdot (q-1), \frac{1}{2} \pm \frac{1}{2q}\right)\right) \cdot \mathrm{P}\left(B\left(\ell, \frac{1}{2} + \delta\right) = i\right).$$

Now, we prove that for a big enough modulo q, Algorithm 5 is successful:

Lemma 3. *Let C, T, δ be as specified in Algorithm 5. Then, if $\pm\frac{1}{2q} < \delta$, then the above algorithm is successful.*

Proof. Since by hypothesis, both distributions, namely, uniform and PLWE, are equiprobable, we need to prove that the following holds:

$$\frac{1}{2} \cdot (\mathrm{P}(C < T | D = U) + \mathrm{P}(C \geq T | D = \mathcal{G}_\sigma)) = \frac{1}{2} + \frac{1}{2} \cdot (\mathrm{P}(C < T | D = U) - \mathrm{P}(C < T | D = \mathcal{G}_\sigma)) > \frac{1}{2}.$$

To do so, we will exploit the fact that, if $\frac{1}{2} \pm \frac{1}{2q}$ is close enough to $\frac{1}{2}$, we get the result.

We compute the second term in the subtraction in terms of the number C_g for all $g \neq g^*$, reasoning as above:

$$\begin{aligned}\mathrm{P}(C < T | D = \mathcal{G}_\sigma) &= \sum_{i=0}^{\ell q - \ell} \mathrm{P}(C - C_g < T - i | D = \mathcal{G}_\sigma) \cdot \mathrm{P}(C_g = i) \\ &= \sum_{i=0}^{\ell q - \ell} F\left(T - i - 1, \ell, \frac{1}{2} + \delta\right) \cdot \mathrm{P}\left(B\left(\ell q - \ell, \frac{1}{2} \pm \frac{1}{2q}\right) = i\right).\end{aligned}$$

Now for the first term we proceed similarly, keeping in mind that we are in the case of the uniform distribution:

$$\mathrm{P}(C < T | D = U) = \sum_{i=0}^{\ell q - \ell} F\left(T - i - 1, \ell, \frac{1}{2} \pm \frac{1}{2q}\right) \cdot \mathrm{P}\left(B\left(\ell q - \ell, \frac{1}{2} \pm \frac{1}{2q}\right) = i\right).$$

Exploiting the decreasing monotonicity of the cumulative binomial present in the first term and the fact that the second term is the same in both probabilities, we conclude that Algorithm 5 is successful if

$$\Delta := \frac{1}{2} + \delta - \left(\frac{1}{2} \pm \frac{1}{2q}\right) = \delta - \left(\pm\frac{1}{2q}\right) > 0.$$

□

Success probability depending on the 'distribution ratio'. The probability of success is depending on the formula laid out above, which is based on the following computations:

1. $\int_0^\infty \mathcal{G}_{\overline{\sigma}} = \frac{1}{2}$
2. $\int_0^{\frac{q}{4}} \mathcal{G}_{\overline{\sigma}} = \frac{1}{2} \operatorname{erf}\left(\frac{q}{4\sqrt{2} \cdot \overline{\sigma}}\right)$
3. $\int_{\frac{3 \cdot q}{4} + j \cdot q}^{\frac{5 \cdot q}{4} + j \cdot q} \mathcal{G}_{\overline{\sigma}} = \frac{1}{2}\left(\operatorname{erf}\left(q \frac{5/4 + j}{\sqrt{2} \cdot \overline{\sigma}}\right) - \operatorname{erf}\left(q \frac{3/4 + j}{\sqrt{2} \cdot \overline{\sigma}}\right)\right)$

where $\operatorname{erf}(x) = \frac{\sqrt{2}}{\pi} \cdot \int_0^x e^{-t^2} \cdot dt$ is the 'error function'. A common value, $r = \frac{q}{\sqrt{2} \cdot \overline{\sigma}}$, appears. This value, referred to from now on as the *distribution ratio*, reigns over the probability value and therefore its convergence will be analyzed based on it.

Since our assumption is $2\overline{\sigma} \geq \frac{q}{4}$, the *distribution ratio* satisfies $r \in (0, 4\sqrt{2}]$. Therefore, plotting the function

$$f(r) = \operatorname{erf}(r/4) + \sum_{j \in \mathbb{N}} (\operatorname{erf}(r \cdot (5/4 + j)) - \operatorname{erf}(r \cdot (3/4 + j)))$$

and calculating when the condition $f(r) > \frac{1}{2} \pm \frac{1}{2q}$ is satisfied, for values of r in the interval $(0, 4\sqrt{2}]$, completely characterizes the PLWE values that give rise to a successful attack.

Practical instances. We provide a number of *toy* examples to demonstrate the applicability of the *unbounded small values attack*:

1. – $N = 256$
   – $q = 3329$.
   – $\alpha = 3330$, order of α in $\mathbb{F}_q^*$ is $r = 2$.
   – $\sigma = 8$.
   – $\mathrm{P}(E_i | D = \mathcal{G}_\sigma) \approx 0.5983462$.
   – $\Delta \approx 0.0981960$
2. – $N = 256$
   – $q = 3677$.
   – $\alpha = 3676$, order of α in $\mathbb{F}_q^*$ is $r = 2$.
   – $\sigma = 8$.
   – $\mathrm{P}(E_i | D = \mathcal{G}_\sigma) \approx 0.6377288$.
   – $\Delta \approx 0.13759$
3. – $N = 256$
   – $q = 2887$.
   – $\alpha = 698$, order of α in $\mathbb{F}_q^*$ is $r = 3$.
   – $\sigma = 8$.
   – $\mathrm{P}(E_i | D = \mathcal{G}_\sigma) \approx 0.4999999$.
   – $\Delta \approx 0.00017$
4. – $N = 256$
   – $q = 4111$.
   – $\alpha = 1055$, order of α in $\mathbb{F}_q^*$ is $r = 3$.
   – $\sigma = 8$.
   – $\mathrm{P}(E_i | D = \mathcal{G}_\sigma) \approx 0.5000000000009331$.
   – $\Delta \approx 0.0001216$

Regarding application of this attack to practical known instances deployed, the situation is less likely to be fruitful.

Current PLWE schemes are mostly based upon cyclotomic rings, i.e., polynomial rings constructed from a cyclotomic polynomial. The specific nature of these polynomials (i.e., the fact that their roots are primitive roots of unity) causes that the order associated with their roots (the *conductor/index* of the cyclotomic polynomial) will be high enough so that the attack cannot be computed, due to the size of the value of the resulting standard deviation of the image Gaussian distribution.

Note that while this will remain the case over finite extensions of degree n of $\mathbb{F}_q$ (see Section 5) as the elements considered there in the cyclotomic case will also be $\frac{N}{gcd(n,N)}$-th primitive roots of unity, the use of higher-degree extension can make the attacks more applicable, as both the order of the elements and the powers involved decrease.

4 Small Set Attack for roots over finite field extensions of $\mathbb{F}_q$

In [1] and [3], an attack was presented when there existed a root α of $f(x)$ belonging to a proper $\mathbb{F}_q$ quadratic extension. The idea of that attack was to replace the set of the likely values for $e(\alpha)$ by the set of their $\mathbb{F}_q$-traces, hence drastically reducing the size of the small set at the cost of producing just a decisional attack, instead of a search one.

However, the attack just described suggests a natural extension to settings where $f(x)$ has a higher-than-quadratic-degree factor. This is actually what we will do in the coming Subsections for the *small set attack*, and in Section 5 for the *small values attack*.

Next we revisit traces and also introduce some tools, which will prove important for the present and the coming Sections.

4.1 Preliminaries

Recall that R_q is the ring $\mathbb{F}_q[x]/(f(x))$, with $f(x)$ a monic irreducible polynomial over $\mathbb{Z}[x]$ of degree N. To introduce this new setting, suppose now that the following conditions are met:

– There exists $a \in \mathbb{F}_q$ such that the polynomial $x^n - a$ divides $f(x)$ in $\mathbb{F}_q[x]$ and, therefore, $n < N$.
– The polynomial $x^n - a$ is irreducible over $\mathbb{F}_q[x]$, i.e., $f(x)$ has an irreducible factor of degree n.
– The element a has a small multiplicative order in $\mathbb{F}_q^*$.

We choose $\alpha \in \mathbb{F}_{q^n} \setminus \mathbb{F}_q$ such that its minimal polynomial over $\mathbb{F}_q$ is precisely $x^n - a$. To pursue the adaptation of the attack described in [1] to the higher-degree setting we define now

$$R_{q,0} = \{p(x) \in R_q : p(\alpha) \in \mathbb{F}_q\}.$$

Then we have the following

Proposition 11. *The set $R_{q,0}$ is a ring and, as an $\mathbb{F}_q$-vector subspace of R_q, has dimension $N-n+1$.*

Proof. The fact that it is a subring of R_q is obvious. As for the dimension we observe that, given $p(x) = \sum_{i=0}^{N-1} p_i x^i$, we have that $p(x) \in R_{q,0}$ if and only if

$$p(\alpha) = \sum_{j=0}^{N'-1} a^j p_{nj} + \alpha \sum_{j=0}^{N'-1} a^j p_{nj+1} + \cdots + \alpha^{n-1} \sum_{j=0}^{N'-1} a^j p_{nj+(n-1)} \in \mathbb{F}_q,$$

with $N' = \frac{N}{n}$, assuming for simplicity that $n \mid N$. Hence $p(\alpha) \in \mathbb{F}_q$ if and only if

$$\sum_{j=0}^{N'-1} a^j p_{nj+1} = 0,$$
$$\sum_{j=0}^{N'-1} a^j p_{nj+2} = 0,$$
$$\cdots$$
$$\sum_{j=0}^{N'-1} a^j p_{nj+n-1} = 0,$$

and thus the result follows. □

We introduce now traces in this setting and show some interesting properties. Let us begin with the following

Proposition 12. *Suppose that $x^n - a$ divides $f(x)$ over $\mathbb{F}_q[x]$ for some $n < N$. Assume that $x^n - a$ is irreducible over $\mathbb{F}_q[x]$ and let $\alpha \in \mathbb{F}_{q^n} \setminus \mathbb{F}_q$ be a root of $f(x)$. Then, for each $j \geq 1$ such that j is not a multiple of n we have that $\mathrm{Tr}(\alpha^j) = 0$.*

Proof. First, assume that $1 \leq j < n$ and let $\alpha_1, \ldots, \alpha_n$ be the conjugates of $\alpha_1 := \alpha$ in $\mathbb{F}_{q^n}$. We have

$$\mathrm{Tr}(\alpha^j) = \sum_{j=1}^{n} \alpha_i^j,$$

which is a symmetric expression evaluated at the conjugated roots. Moreover, the symmetric polynomial $\mathrm{Tr}_j(x_1, \ldots, x_n) := \sum_{i=1}^n x_i^j$ is homogeneous of degree j. Hence, we can express

$$\mathrm{Tr}_j(x_1, \ldots, x_n) = g(s_1, \ldots, s_n),$$

where $g(x_1, \ldots, x_n)$ is some polynomial and $s_i(x_1, \ldots, x_n)$ is the i-th elementary symmetric polynomial in the indeterminates $x_1, \ldots, x_n$ (see [21, 1.12]). Since the Tr_j has no independent term we have that $g(0, \ldots, 0) = 0$ and moreover, since $j < n$, $g(x_1, \ldots, x_n)$ does not contain monomial terms divisible by x_n. Finally, again by Cardano-Vieta, since all the coefficients of $x^n - a$ vanish except for the first and the last, we have that $s_i(\alpha_1, \ldots, \alpha_n) = 0$ for $1 \leq i < n$ so that $\mathrm{Tr}(\alpha^j) = 0$.

Now, for $j > n$ not a multiple of n we can write $j = tn + k$ with $1 \leq k < n$ and hence

$$\mathrm{Tr}(\alpha^j) = a^t \,\mathrm{Tr}(\alpha^k) = 0.$$

□

Another important feature of the trace operator is that it respects the distribution of the candidate values that will be considered under the attacks:

Proposition 13. *The trace operator respects the distribution of the tentative error values* $b(\alpha)-g\cdot a(\alpha)$*, for any fixed* $g \in \mathbb{F}_{q^n}$*.*

Proof. It is required to see that the traces of the evaluated tentative error polynomials follow the same distributions as the case $\alpha \in \mathbb{F}_q$.

Define $a(x) = \sum_{i=0}^{N-1} a_i x^i$, $b(x) = \sum_{i=0}^{N-1} b_i x^i$. If the samples are uniform, i.e., a_i, b_i are uniform in $\mathbb{F}_q$ then, given $g \in \mathbb{F}_{q^n}$,

$$\mathrm{Tr}(b(\alpha) - g \cdot a(\alpha)) = \sum_{j=0}^{N-1} b_j \cdot \mathrm{Tr}(\alpha^j) - \mathrm{Tr}\left(\sum_{j=0}^{N-1} a_j \cdot \alpha^j \cdot g\right) = \sum_{j=0}^{N-1} b_j \cdot \mathrm{Tr}(\alpha^j) - \sum_{j=0}^{N-1} a_j \cdot \mathrm{Tr}(\alpha^j \cdot g),$$

which is the sum of two linear combination of independent uniform distributions in $\mathbb{F}_q$ and coefficients in $\mathbb{F}_q$ and, hence, by Lemma 1, is a uniform distribution in $\mathbb{F}_q$.

Define $g^* = s(\alpha)$. If the samples are PLWE, then for the case $g = g^*$, we have:

$$\mathrm{Tr}(b(\alpha) - g^* \cdot a(\alpha)) = \mathrm{Tr}(e(\alpha)) = \sum_{j=0}^{N-1} e_j \cdot \mathrm{Tr}(\alpha^j),$$

which is the weighted (by $\mathrm{Tr}(\alpha^j) \in \mathbb{F}_q$) sum of (at most) N Gaussian random variables $e_j \in \mathbb{F}_q$ of mean 0 and variance σ^2 and, therefore, is a Gaussian distribution of mean 0 and a certain variance $\overline{\sigma}^2$ in $\mathbb{F}_q$.

Conversely, for the case $g \neq g^*$,

$$\mathrm{Tr}(b(\alpha) - g \cdot a(\alpha)) = \mathrm{Tr}(e(\alpha) + (s(\alpha) - g) \cdot a(\alpha)) = \mathrm{Tr}(e(\alpha)) + \sum_{j=0}^{N} a_j \cdot \mathrm{Tr}(\alpha^j \cdot (s(\alpha) - g)),$$

which is the sum of a Gaussian distribution of mean 0 and a variance $\overline{\sigma}^2$ and a uniform distribution in $\mathbb{F}_q$, and therefore the result follows. □

4.2 Layout of the generalized attack

We present now the generalized attack divided into two stages, each one of them giving rise to a specific algorithm.

The first stage is carried out by Algorithm 6, whose input is a set of samples S coming from $R_{q,0} \times R_q$ and allowing one to distinguish whether such set is drawn either from the uniform or from the PLWE distribution. In particular, the outcome of the Algorithm will be one out of three possible results: **PLWE**, which means that the samples come from a PLWE distribution; **NOT PLWE**, which means the samples come from the uniform distribution; or **NOT ENOUGH SAMPLES**, which means that the number of samples in the set is not enough to provide a definitive conclusion.

Observe, then, that if $(a(x), b(x) = a(x)s(x) + e(x)) \in R_{q,0} \times R_q$ is a PLWE sample, evaluating at α we have that $\mathrm{Tr}(b(\alpha) - a(\alpha)s(\alpha)) = \mathrm{Tr}(e(\alpha))$. Computing the latter

$$\mathrm{Tr}(e(\alpha)) = \sum_{i=0}^{N-1} e_i \cdot \mathrm{Tr}(\alpha^i) = \sum_{j=0}^{N'-1} e_{nj} \cdot \mathrm{Tr}(\alpha^{nj}) = n \cdot \sum_{j=0}^{N'-1} e_{nj} \cdot a^j, \tag{4}$$

where $N' = \frac{N}{n}$ and, for simplicity, $n \mid N$. Assuming that r is the multiplicative order of a in $\mathbb{F}_q^*$, let us define $N'' = \frac{N'}{r}$ and, again for simplicity, suppose that $r \mid N'$. Thus we eventually obtain

$$\mathrm{Tr}(e(\alpha)) = n \sum_{i=0}^{r-1} a^i \cdot \sum_{j=0}^{N''-1} e_{n(jr+i)}. \tag{5}$$

Now, two remarks are in order.

Input: A set of M samples $S = \{(a_i(x), b_i(x))\}_{i=1}^M \in R_{q,0} \times R_q$
A look-up table Σ of values appearing in $\frac{1}{n}\operatorname{Tr}(b(\alpha) - a(\alpha)\cdot s(\alpha))$ with probability $\geq p_0^r$
Output: **PLWE**,
or **NOT PLWE**,
or **NOT ENOUGH SAMPLES**

- $G := \emptyset$
- ***for*** $g \in \mathbb{F}_q$ ***do***
  - ***for*** $(a_i(x), b_i(x)) \in S$ ***do***
    - ***if*** $\frac{1}{n}(\operatorname{Tr}(b_i(\alpha)) - a_i(\alpha)g) \notin \Sigma$ ***then next*** g
  - $G := G \cup \{g\}$
- ***if*** $G = \emptyset$ ***then return*** **NOT PLWE**
- ***if*** $|G| = 1$ ***then return*** **PLWE**
- ***if*** $|G| > 1$ ***then return*** **NOT ENOUGH SAMPLES**

Algorithm 6. Attack based on the size of the set of possible error values over higher-degree extensions on $R_{q,0}$

1. Since $a(x) \in R_{q,0}$, this means that $a(\alpha) \in \mathbb{F}_q$, by definition. Then
$$\operatorname{Tr}(b(\alpha) - a(\alpha)\cdot s(\alpha)) = \operatorname{Tr}(b(\alpha)) - a(\alpha)\operatorname{Tr}(s(\alpha)).$$
Remark that in the previous expression the only unknown is precisely $\operatorname{Tr}(s(\alpha)) \in \mathbb{F}_q$. This fact allows our algorithm to loop over $\mathbb{F}_q$ rather than over the full ring R_q.
2. For fixed values n and r, let us define $e_i^* = \sum_{j=0}^{N''-1} e_{n(jr+i)}$, $0 \leq i \leq r-1$, so that Equation (5) now reads
$$\operatorname{Tr}(e(\alpha)) = n\sum_{i=0}^{r-1} a^i e_i^*. \tag{6}$$
Remark that each sum e_i^* of coefficients can be seen as a sample from a centered discrete Gaussian of standard deviation less than or equal to $\sqrt{N''}\sigma$ and, henceforth, we can list those elements that occur with probability beyond p_0, which are at most $4\sqrt{N''}\sigma + 1$.

With these at hand, we can construct a look-up table Σ for all the possible values in Equation (5) that happen with probability beyond p_0^r. Such look-up table is defined as follows:
$$\Sigma := \left\{\sum_{i=0}^{r-1} a^i x_i \,:\, |x_i| \leq 2\sqrt{N''}\sigma, x_i \in \mathbb{Z}\right\}.$$
Observe that
$$|\Sigma| \leq \left(4\sqrt{N''}\sigma + 1\right)^r. \tag{7}$$

The second stage of the attack consists essentially in sampling from the oracle presented to the adversary until they obtain a suitable number of samples from $R_{q,0} \times R_q$. We provide Algorithm 7 in order to achieve this task.

Algorithm 7 returns a pair $(a(x), b(x)) \in R_{q,0} \times R_q$ by repeatedly sampling from the original distribution X keeping also track of the number of invocations to X. Identifying this algorithm with a random variable X_0 with values on $R_{q,0} \times R_q$, it is proved in [1] that X is uniform (resp. PLWE) if and only if X_0 is uniform (resp. PLWE).

Notice that the variable *count* stores the expected number of times one has to run X before succeeding. Based on a similar statement found in [1], we have now the next

Proposition 14. *Notations as before, the variable count is distributed as a geometric random variable of first kind with success probability $q^{-(n-1)}$. In particular the expected number of times is just its mean $E[count] = q^{n-1}$*

Proof. Since $\dim_{\mathbb{F}_q}(R_q) = N$ and $\dim_{\mathbb{F}_q}(R_{q,0}) = N - n + 1$, then the probability that an element taken uniformly from R_q^2 actually belongs to $R_{q,0} \times R_q$ is precisely $q^{N-n+1}/q^N = q^{1-n}$. Now the average of a geometric random variable of the first kind of success probability θ is $\frac{1-\theta}{\theta}$ which is our case equals $q^{n-1} - 1$ (see for instance [5, Appendix A]). $\square$

Input: A distribution X over R_q^2
Output: A sample $(a(x), b(x)) \in R_{q,0} \times R_q$

- $count := 0$
- ***do***
  - $(a(x), b(x)) \xleftarrow{X} R_q^2$
  - $count := count + 1$
- ***until*** $a(x) \in R_{q,0}$
- ***return*** $(a(x), b(x)), count$

Algorithm 7. Random variable X_0

Since, as recommended, q should be of order $\mathcal{O}(N^2)$ for the PLWE cryptosystem to be feasible, the expected number of calls to PLWE oracle to grant that our attack succeeds is of order $\mathcal{O}(N^{2(n-1)})$.

Note that the setting presented allows, theoretically, to direct the laid-out attacks to arbitrary-degree extensions without any further restrictions. Despite this fact, it is clear that the attacks just described rely heavily on the degree of the extension and become almost infeasible for n higher than 4.

4.3 Analysis of the algorithm

The following results establish the reliability of Algorithm 6, over any arbitrary degree extension. We will visit in turn the cases where the Gaussian distributions are either truncated to width 2σ or not truncated.

Let us begin with the case where Gaussian distributions are truncated. Then, we have the following

Proposition 15. *Let Σ, M and r be defined as above, and assume $|\Sigma| < q$. Then:*

1. *If Algorithm 6 returns* ***PLWE*** *or* ***NOT ENOUGH SAMPLES****, then the samples are* ***PLWE*** *with probability at least* $1 - q\left(\frac{|\Sigma|}{q}\right)^M$.
2. *If Algorithm 6 returns* ***NOT PLWE****, then the samples are uniform with probability* 1.

Proof. Let us prove both items in turn.

1. Denote with E the event "Algorithm 6 returns **PLWE** or **NOT ENOUGH SAMPLES**". Applying Bayes' theorem,

$$\mathrm{P}(D = \mathcal{G}_\sigma | E) = 1 - \mathrm{P}(D = U|E) = 1 - \frac{\mathrm{P}(E|D = U)}{\mathrm{P}(E|D = U) + \mathrm{P}(E|D = \mathcal{G}_\sigma)} \geq 1 - \frac{\mathrm{P}(E|D = U)}{\mathrm{P}(E|D = \mathcal{G}_\sigma)}.$$

Let U_g be the event that $\frac{1}{n}\operatorname{Tr}(b_i(\alpha) - a_i(\alpha)g) \in \Sigma$, for all $i = 1, \ldots, M$, i.e., for all samples in S. Then

$$\mathrm{P}(E|D = \mathcal{G}_\sigma) = \mathrm{P}\left(\bigcup_{g=0}^{q-1} U_g\right) \geq \mathrm{P}(U_{g^*}),$$

where $g^* = s(\alpha)$. Observe that $b_i(\alpha) - a_i(\alpha)g^* = e_i(\alpha)$. Since we are dealing with the truncated case, we have

$$\mathrm{P}(U_{g^*}) = \prod_{i=1}^{M} \mathrm{P}(\frac{1}{n}\operatorname{Tr}(e_i(\alpha)) \in \Sigma) = 1,$$

because, according to Equation 6,

$$\frac{1}{n}\operatorname{Tr}(e_i(\alpha)) = \sum_{j=0}^{r-1} a^j \cdot e_j^* \quad \in \Sigma, \quad \forall i$$

and each e_j^* is a (truncated) Gaussian random variable.
Moreover, $\mathrm{P}(E|D=U)$ has the same bounds as in the proof of Proposition 1, since $\frac{1}{n}\operatorname{Tr}(b_i(\alpha)-a_i(\alpha)g)$ is uniformly random in $\mathbb{F}_q$ by Proposition 13. Combining results,

$$\mathrm{P}(D=\mathcal{G}_\sigma|E)\geq 1-\frac{\mathrm{P}(E|D=U)}{\mathrm{P}(E|D=\mathcal{G}_\sigma)}=1-\mathrm{P}(E|D=U)\geq 1-q\left(\frac{|\Sigma|}{q}\right)^M.$$

2. Denote again with E' the event "Algorithm 6 returns **NOT PLWE**". Applying Bayes' theorem,

$$\mathrm{P}(D=U|E')=\frac{\mathrm{P}(E'|D=U)}{\mathrm{P}(E'|D=U)+\mathrm{P}(E'|D=\mathcal{G}_\sigma)}.$$

Remark as before that the event E' is complementary to the event E, defined in the previous item. Hence

$$\mathrm{P}(E'|D=\mathcal{G}_\sigma)=1-\mathrm{P}(E|D=\mathcal{G}_\sigma)=0.$$

Consequently,

$$\mathrm{P}(D=U|E')=1.$$

□

We turn now our attention to the non-truncated case, by stating the following

Proposition 16. *Let Σ, M and r be defined as above, and assume $|\Sigma|<qp_0^r$. Then:*

1. *If Algorithm 6 returns **PLWE** or **NOT ENOUGH SAMPLES**, then the samples are **PLWE** with probability at least $1-q\left(\frac{|\Sigma|}{qp_0^r}\right)^M$.*
2. *If Algorithm 6 returns **NOT PLWE**, then the samples are uniform with probability at least $1/2$ for large enough M.*

Proof. Let us prove both items in turn.

1. As before, denote with E the event "Algorithm 6 returns **PLWE** or **NOT ENOUGH SAMPLES**". Applying again Bayes' theorem,

$$\mathrm{P}(D=\mathcal{G}_\sigma|E)\geq 1-\frac{\mathrm{P}(E|D=U)}{\mathrm{P}(E|D=\mathcal{G}_\sigma)}.$$

Let U_g be the event that $\frac{1}{n}\operatorname{Tr}(b_i(\alpha)-a_i(\alpha)g)\in\Sigma$, for all $i=1,\ldots,M$, i.e., for all samples in S. Then

$$\mathrm{P}(E|D=\mathcal{G}_\sigma)=\mathrm{P}\left(\bigcup_{g=0}^{q-1}U_g\right)\geq\mathrm{P}(U_{g^*}),$$

where $g^*=s(\alpha)$. Since we are now dealing with the not truncated case, we have

$$\mathrm{P}(U_{g^*})=\prod_{i=1}^{M}\mathrm{P}(e_i(\alpha)\in\Sigma)\geq p_0^{Mr}.$$

Moreover, $\mathrm{P}(E|D=U)$ is the same as in the truncated case, since the uniform distribution is not affected by the truncation. Combining results,

$$\mathrm{P}(D=\mathcal{G}_\sigma|E)\geq 1-\frac{\mathrm{P}(E|D=U)}{\mathrm{P}(E|D=\mathcal{G}_\sigma)}\geq 1-\frac{\mathrm{P}(E|D=U)}{p_0^{Mr}}\geq 1-\frac{q\left(\frac{|\Sigma|}{q}\right)^M}{p_0^{Mr}}=1-q\left(\frac{|\Sigma|}{qp_0^r}\right)^M.$$

2. Denote again with E' the event "Algorithm 6 returns **NOT PLWE**". Applying Bayes' theorem,

$$\mathrm{P}(D=U|E')=\frac{\mathrm{P}(E'|D=U)}{\mathrm{P}(E'|D=U)+\mathrm{P}(E'|D=\mathcal{G}_\sigma)}.$$

Remark as before that the event E' is complementary to the event E, defined in the previous item. Hence

$$\mathrm{P}(E'|D=\mathcal{G}_\sigma)=1-\mathrm{P}(E|D=\mathcal{G}_\sigma)\leq 1-p_0^{Mr}.$$

Consequently,

$$\mathrm{P}(D = U|E') \geq \frac{\mathrm{P}(E'|D = U)}{\mathrm{P}(E'|D = U) + 1 - p_0^{Mr}},$$

which is greater than $1/2$ if and only if $\mathrm{P}(E'|D = U) \geq 1 - p_0^{Mr}$. Since

$$\mathrm{P}(E'|D = U) \geq 1 - q\left(\frac{|\Sigma|}{q}\right)^M,$$

for the conclusion to hold it is enough to take M such that $q^{\frac{1}{M}}\frac{|\Sigma|}{q} \leq p_0^r$.

□

All told, whenever $\left(4\sqrt{N''}\sigma + 1\right)^r < qp_0^r$, we have a decisional attack against the $R_{q,0} \times R_q$ PLWE problem. This condition already imposes a restriction on n, since $r \leq q^n - 1$ so that for the look-up table to be checked in feasible time at each iteration, we must restrict to roots such that r is *small enough* in relation to n, which is difficult to test unless n has a suitable size. This is one of the reasons why we focus in the case $n = 2, 3$ or at most 4. Nevertheless, from a theoretical point of view, the presented generalization does not impose restrictions on n.

Success probabilities. Observe that from the discussion above, we can derive the success probability of Algorithm 6. For the truncated case, we have the following

Proposition 17. *With the same notations and assumptions as in Proposition 15,*

1. *If the samples are PLWE, Algorithm 6 guesses correctly with probability 1*
2. *If the samples are uniform, Algorithm 6 guesses correctly with probability at least* $1 - q\left(\frac{|\Sigma|}{q}\right)^M$.

Proof. The first item is equivalent to computing $\mathrm{P}(E|D = \mathcal{G}_\sigma)$, which is 1, according to the proof in Proposition 15. The second item is not affected by the truncation so the bound is the same as for the truncated case, namely, greater or equal to $1 - q\left(\frac{|\Sigma|}{q}\right)^M$. □

For the not truncated case, we have the following

Proposition 18. *With the same notations and assumptions as in Proposition 16,*

1. *If the samples are PLWE, Algorithm 6 guesses correctly with probability at least* p_0^{Mr}.
2. *If the samples are uniform, Algorithm 6 guesses correctly with probability at least* $1 - q\left(\frac{|\Sigma|}{q}\right)^M$.

Proof. The first item is equivalent to computing $\mathrm{P}(E|D = \mathcal{G}_\sigma)$, which is at least p_0^{Mr}, according to the proof in Proposition 16. The second item is not affected by the truncation so the bound is the same as for the truncated case, namely, greater or equal to $1 - q\left(\frac{|\Sigma|}{q}\right)^M$. □

An improvement for a higher number of samples. As discussed in Section 3.1, the extension algorithm presented, which employed Algorithm 1 as a sub-process, did not employ any feature specific of $\alpha \in \mathbb{F}_q$. As such, the same discussion and results presented there can be applied to finite extensions of $\mathbb{F}_q$, giving rise to the following algorithm:

Input:	A collection of samples $S = \{(a_i(x), b_i(x))\}_{i=1}^{M} \in R_{q,0} \times R_q$
	A choice M_0 for the number of samples of the sub-process
Output:	A guess for the distribution of the samples, either
	PLWE or **UNIFORM**

- $T := \lceil \lfloor M/M_0 \rfloor \cdot p_0^{M_0 r} \rceil$
- $C := 0$
- ***for*** $j \in \{0, ..., \lfloor M/M_0 \rfloor - 1\}$ ***do***
  - *result* $:= \mathrm{SmallSetAttackTraces}\left(S_j := \{(a_i(x), b_i(x))\}_{i=j\cdot M_0+1}^{(j+1)\cdot M_0}\right)$
    - ***if*** *result* $\neq$ **NOT PLWE** ***then***
      - $C := C + 1$
- ***if*** $C < T$ ***then return*** **UNIFORM**
- ***else return*** **PLWE**

Algorithm 8. Algorithm for Extended Small Set Attack over finite extensions of $\mathbb{F}_q$

Moreover, since the success probabilities detailed in Proposition 18 for Algorithm 6 are the same as the ones in Proposition 4 for Algorithm 1, all the results detailed in Section 3.1 regarding the success probabilities of the extended attack apply here as well.

In particular, this means that:

- Case $D = \mathcal{G}_\sigma$: We have that

$$\mathrm{P}(C \geq T | D = \mathcal{G}_\sigma) := 1 - F(T-1, \lfloor M/M_0 \rfloor, \mathrm{P}_{\mathcal{G}_\sigma}(M_0)) \geq 1 - F(T-1, \lfloor M/M_0 \rfloor, p_0^{M_0 r})$$

- Case $D = U$: We have:

$$\mathrm{P}(C < T | D = U) := F(T-1, \lfloor M/M_0 \rfloor, \mathrm{P}_U(M_0)) \geq F(T-1, \lfloor M/M_0 \rfloor, 1 - (|\Sigma|/q)^{M_0})$$

Proposition 19. *Let C, T, M_0 be as specified in Algorithm 8. Then, if $1 - (|\Sigma|/q)^{M_0} < p_0^{M_0 r}$, Algorithm 8 is successful.*

Proof. See proof of Proposition 5. □

Practical instances. We evaluated the trace attack generalization over two *toy* instances to exemplify the practical applications of the above presented attacks. We have:

1. - $f = x^{23} - 2018x^{20} + x^{13} - 2018x^{10} + 2017x^3 + 1$.
   - $q = 4099$.
   - $a = 2018$, order of a in $\mathbb{F}_q^*$ is $r = 6$.
   - $g(x) = x^3 - 2018$ irreducible in $\mathbb{F}_q[x]$ and a divisor of $f(x)$.
   - $N' = 7$, $N'' = 1$.
   - $\sigma = 0.7$.
   - $\left\lfloor 4\sqrt{N''}\sigma + 1 \right\rfloor^r \approx 3010 < q p_0^r$.

   Using these values, if Algorithm 6 returns **PLWE**, then the samples are indeed **PLWE** with probability 0.861 for 350 samples, and with probability 0.998 for 500 samples.
2. - $f(x) = x^{23} - 2017x^{20} + x^{13} - 2017x^{10} + 2018x^3 + 1$.
   - $q = 4099$.
   - $a = 2017$, order of a in $\mathbb{F}_q^*$ is $r = 3$.
   - $g(x) = x^3 - 2017$ irreducible in $\mathbb{F}_q[x]$ and a divisor of $f(x)$.
   - $N' = 7$, $N'' = 2$.
   - $\sigma = 5/2$.
   - $\left(4\sqrt{N''}\sigma + 1\right)^r \approx 3471 < q p_0^r$.

   For this second case, if Algorithm 6 returns **PLWE**, then the samples are indeed **PLWE** with probability 0.629 for 350 samples, and with probability 0.993 for 500 samples.

Notice that these simulations require less than one half of the number of samples than in [8] and, moreover, we do not require the roots to belong to $\mathbb{F}_q$. On the other hand, we are far from the improvement given in [6] but it is convenient to point out that their attack only applies to two particular (although infinite) family of trinomials.

We conducted also experiments to check the ability to find elements in $R_{q,0}$. The details can be found in Appendix A.

5 Small Values Attack for roots over finite extensions of $\mathbb{F}_q$

In the previous section, we introduced a way to generalize root-based attacks, focused on the *small set attack* developed in [13]. In this section, we translate the same ideas to the *Attack based on the size of error values*, and provide with a generalization of these attacks for arbitrary degree extensions of $\mathbb{F}_q$, exploiting the trace setting as in the previous sections. As such, the general framework employed in the previous section will apply here as well.

5.1 Preliminaries

Within this setting, we apply Proposition 12. Therefore, given the polynomial $x^n - a$, we have:

1. If i is not a multiple of n, $\mathrm{Tr}(\alpha^i) = 0$.
2. If i is of the form $n \cdot j$, then $\mathrm{Tr}(\alpha^i) = \mathrm{Tr}(a^j) = a^j \cdot \mathrm{Tr}(1) = n \cdot a^j$.

The cases to consider are as follows, given the independent term a of the polynomial that builds the n-degree extension:

1. Case $a = \pm 1$.
2. Case $a \neq \pm 1$ and has small order r modulo q.
3. Case $a \neq \pm 1$ and does not have small order modulo q.

When migrating to the trace setting under this attack, we have to study whether the traces of the error values, once evaluated in α and taken modulo q, are small. For convenience, let us remember Equation (4) where the trace of an error value is presented:

$$\mathrm{Tr}(e(\alpha)) = \sum_{i=0}^{N-1} e_i \cdot \mathrm{Tr}(\alpha^i) = \sum_{j=0}^{N'-1} e_{nj} \cdot \mathrm{Tr}(\alpha^{nj}) = n \cdot \sum_{j=0}^{N'-1} e_{nj} \cdot a^j,$$

where $N' = \frac{N}{n}$ and, for simplicity, $n \mid N$. Now we deal in turn with the cases above:

Case $a = \pm 1$. In this case, the value $\mathrm{Tr}(e(\alpha)) = n \sum_{j=0}^{N'-1} e_{nj} \cdot a^j$ is (n times) a sum of N' Gaussians of mean 0 and variance σ^2, so it is a Gaussian of mean 0 and variance $N' \cdot \sigma^2$. Therefore $\mathcal{G}_{\overline{\sigma}}$ is a Gaussian with $\overline{\sigma}^2 = N' \cdot \sigma^2$.

Case $a \neq \pm 1$ and has small order r modulo q. Here, the small order property of the element a is used to group the error value $\mathrm{Tr}(e(\alpha))$ into r packs of $\frac{N'}{r}$ values (assuming for simplicity that $r \mid N'$). Each pack forms an independent Gaussian of mean 0 and variance $\frac{N'}{r} \cdot \sigma^2$, and thus the overall distribution is a weighted sum of such Gaussians, which generates a Gaussian of mean 0 and variance $\overline{\sigma}^2 = \sum_{i=0}^{r-1} \frac{N'}{r} \cdot \sigma^2 \cdot a^{2 \cdot i}$.

Case $a \neq \pm 1$ and does not have small order modulo q. In the most general case, we just have a weighted sum of N' Gaussians, which is itself a Gaussian of mean 0 and variance $\overline{\sigma}^2 = \sum_{i=0}^{N'-1} \sigma^2 \cdot a^{2 \cdot i}$.

5.2 Small Values Attack

Now, regardless of the above cases, the argument goes as follows: we will work with the values of the Gaussian image generated by the trace of the evaluation of the error polynomials inside $[-2\overline{\sigma}, 2\overline{\sigma}]$, which have a combined mass probability of p_0. If $2\overline{\sigma} < \frac{q}{4}$ is satisfied, Algorithm 9 can be executed.

The introduction of higher-degree extensions on this attack not only allows to generalize the setting in which this attack may be applicable, increasing the reach of it, but also makes it more likely that the necessary condition $2\overline{\sigma} < \frac{q}{4}$ holds.

Actually, introducing the trace creates a narrower $\overline{\sigma}$ value, due to the fact that fewer error terms are considered (most trace values are 0), and the fact that the order of the value a is a divisor of the order of the root α.

In turn, the above attack severely increases the computational complexity of the attack, on two fronts: The main loop over the guesses g is displaced from $\mathbb{F}_q$ to $\mathbb{F}_{q^n}$, and the requirement of evaluating the trace on each sample and guess value $b_i(\alpha) - g \cdot a_i(\alpha)$.

To deal with it, we resort to the auxiliary ring $R_{q,0}$ that was introduced in [1] and we have already used in Subsection 4.1. With this modification, the two-fold increase in computational complexity is alleviated, as the loop can now be restored to $\mathbb{F}_q$, and the trace function need not be evaluated for each sample and guess, it can be pre-computed for every sample before running the algorithm.

Input:	A collection of samples $S = \{(a_i(x), b_i(x))\}_{i=1}^{M} \subseteq R_{q,0} \times R_q$
Output:	A guess $g \in \mathbb{F}_q$ for $\mathrm{Tr}(s(\alpha))$, or **NOT PLWE**, or **NOT ENOUGH SAMPLES**

- $G := \emptyset$
- ***for*** $g \in \mathbb{F}_q$ ***do***
  - ***for*** $(a_i(x), b_i(x)) \in S$ ***do***
    - ***if*** $\frac{1}{n}(\mathrm{Tr}(b_i(\alpha)) - a_i(\alpha)g) \notin [-\frac{q}{4}, \frac{q}{4})$ ***then next*** g
  - $G := G \cup \{g\}$
- ***if*** $G = \emptyset$ ***then return*** **NOT PLWE**
- ***if*** $G = \{g\}$ ***then return*** g
- ***if*** $|G| > 1$ ***then return*** **NOT ENOUGH SAMPLES**

Algorithm 9. Attack based on the size of error values over higher-degree extensions on $R_{q,0}$

5.3 Analysis of the algorithm

As in previous sections, we begin with the analysis of the truncated case:

Proposition 20. *Assume $2\overline{\sigma} < \frac{q}{4}$. Let M be the number of input samples.*

1. *If Algorithm 9 returns **NOT PLWE**, then the samples come from the uniform distribution, with probability at least $1 - q\left(\frac{1}{2} \pm \frac{1}{2q}\right)^M$.*
2. *If it outputs anything other than **NOT PLWE**, then the samples are valid PLWE samples with probability 1.*

Proof. Let us prove both items in turn.

1. As before, denote with E the event "Algorithm 9 returns **PLWE** or **NOT ENOUGH SAMPLES**". Applying again Bayes' theorem,
$$\mathrm{P}(D = \mathcal{G}_\sigma | E) = 1 - \mathrm{P}(D = U|E) = 1 - \frac{\mathrm{P}(E|D = U)}{\mathrm{P}(E|D = U) + \mathrm{P}(E|D = \mathcal{G}_\sigma)} \geq 1 - \frac{\mathrm{P}(E|D = U)}{\mathrm{P}(E|D = \mathcal{G}_\sigma)}.$$
Let U_g be again the event that $\frac{1}{n}\mathrm{Tr}(b_i(\alpha) - a_i(\alpha)g) \in \left[-\frac{q}{4}, \frac{q}{4}\right)$, for all $i = 1, \ldots, M$, i.e., for all samples in S. Then
$$\mathrm{P}(E|D = \mathcal{G}_\sigma) = \mathrm{P}\left(\bigcup_{g=0}^{q-1} U_g\right) \geq \mathrm{P}(U_{g^*}),$$

where $g^* = s(\alpha)$. Observe that $b_i(\alpha) - a_i(\alpha)g^* = e_i(\alpha)$. Since we are dealing with the truncated case and since $[-2\overline{\sigma}, 2\overline{\sigma}) \subseteq \left[-\frac{q}{4}, \frac{q}{4}\right)$ we have

$$\mathrm{P}(U_{g^*}) = \prod_{i=1}^{M} \mathrm{P}\left(\frac{1}{n}\operatorname{Tr}(e_i(\alpha)) \in \left[-\frac{q}{4}, \frac{q}{4}\right)\right) = 1.$$

as $\frac{1}{n}\operatorname{Tr}(e_i(\alpha))$ is a (truncated) Gaussian random variable of mean 0 and variance $\overline{\sigma}^2$. Moreover, $\mathrm{P}(E|D = U)$ has the same bounds as in the proof of Proposition 6, since $\frac{1}{n}\operatorname{Tr}(b_i(\alpha) - a_i(\alpha)g)$ is uniformly random in $\mathbb{F}_q$ by Proposition 13. Combining results,

$$\mathrm{P}(D = \mathcal{G}_\sigma|E) \geq 1 - \frac{\mathrm{P}(E|D = U)}{\mathrm{P}(E|D = \mathcal{G}_\sigma)} = 1 - \mathrm{P}(E|D = U) = 1 - q\left(\frac{1}{2} \pm \frac{1}{2q}\right)^M.$$

2. Denote again with E' the event "Algorithm 1 returns **NOT PLWE**". Applying Bayes' theorem,

$$\mathrm{P}(D = U|E') = \frac{\mathrm{P}(E'|D = U)}{\mathrm{P}(E'|D = U) + \mathrm{P}(E'|D = \mathcal{G}_\sigma)}.$$

Remark as before that the event E' is complementary to the event E, defined in the previous item. Hence

$$\mathrm{P}(E'|D = \mathcal{G}_\sigma) = 1 - \mathrm{P}(E|D = \mathcal{G}_\sigma) = 0.$$

Consequently,

$$\mathrm{P}(D = U|E') = 1.$$

□

Meanwhile, for the non truncated case

Proposition 21. *Assume $2\overline{\sigma} < \frac{q}{4}$ and $\frac{1}{2} \pm \frac{1}{2q} < p_0$ (which is the case for $q > 2$). Let M be the number of input samples.*

1. *If Algorithm 9 returns **NOT PLWE**, then the samples come from the uniform distribution, with probability at least $1 - \frac{q\left(\frac{1}{2} \pm \frac{1}{2q}\right)^M}{p_0^M}$.*
2. *If it outputs anything other than **NOT PLWE**, then the samples are valid PLWE samples with probability at least $1/2$ for large enough M.*

Proof. Let us prove both items in turn.

1. As before, denote with E the event "Algorithm 9 returns **PLWE** or **NOT ENOUGH SAMPLES**". As before,

$$\mathrm{P}(D = \mathcal{G}_\sigma|E) \geq 1 - \frac{\mathrm{P}(E|D = U)}{\mathrm{P}(E|D = \mathcal{G}_\sigma)}.$$

Let U_g be again the event that $\frac{1}{n}\operatorname{Tr}(b_i(\alpha) - a_i(\alpha)g) \in \left[-\frac{q}{4}, \frac{q}{4}\right)$, for all $i = 1, \ldots, M$, i.e., for all samples in S. Then

$$\mathrm{P}(E|D = \mathcal{G}_\sigma) = \mathrm{P}\left(\bigcup_{g=0}^{q-1} U_g\right) \geq \mathrm{P}(U_{g^*}),$$

where $g^* = s(\alpha)$. Since we are now dealing with the not truncated case and since $[-2\overline{\sigma}, 2\overline{\sigma}) \subseteq \left[-\frac{q}{4}, \frac{q}{4}\right)$ we have

$$\mathrm{P}(U_{g^*}) = \prod_{i=1}^{M} \mathrm{P}\left(e_i(\alpha) \in \left[-\frac{q}{4}, \frac{q}{4}\right)\right) \geq p_0^M.$$

Moreover, $\mathrm{P}(E|D = U)$ is the same as in the truncated case, since the uniform distribution is not affected by the truncation. Combining results,

$$\mathrm{P}(D = \mathcal{G}_\sigma|E) \geq 1 - \frac{\mathrm{P}(E|D = U)}{\mathrm{P}(E|D = \mathcal{G}_\sigma)} \geq 1 - \frac{\mathrm{P}(E|D = U)}{p_0^M} \geq 1 - \frac{q\left(\frac{1}{2} \pm \frac{1}{2q}\right)^M}{p_0^M}.$$

2. Denote again with E' the event "Algorithm 9 returns **NOT PLWE**". As before,

$$\mathrm{P}(D=U|E')=\frac{\mathrm{P}(E'|D=U)}{\mathrm{P}(E'|D=U)+\mathrm{P}(E'|D=\mathcal{G}_\sigma)}.$$

Remark as before that the event E' is complementary to the event E, defined in the previous item. Hence

$$\mathrm{P}(E'|D=\mathcal{G}_\sigma)=1-\mathrm{P}(E|D=\mathcal{G}_\sigma)\leq 1-p_0^M.$$

Consequently,

$$\mathrm{P}(D=U|E')\geq\frac{\mathrm{P}(E'|D=U)}{\mathrm{P}(E'|D=U)+1-p_0^M},$$

which is greater than $1/2$ if and only if $\mathrm{P}(E'|D=U)\geq 1-p_0^M$. Since

$$\mathrm{P}(E'|D=U)\geq 1-q\left(\frac{1}{2}\pm\frac{1}{2q}\right)^M,$$

for the conclusion to hold it is enough to take M such that $q^{\frac{1}{M}}\left(\frac{1}{2}\pm\frac{1}{2q}\right)\leq p_0$.

□

Success probabilities. Observe that from the discussion above, we can derive the success probability of Algorithm 9. For the truncated case, we have the following

Proposition 22. *With the same notations and assumptions as in Proposition 20,*

1. *If the samples are PLWE, Algorithm 9 guesses correctly with probability* 1.
2. *If the samples are uniform, Algorithm 9 guesses correctly with probability at least* $1-q(\frac{1}{2}\pm\frac{1}{2q})^M$.

Proof. The first item is equivalent to computing $\mathrm{P}(E|D=\mathcal{G}_\sigma)$, which is equal to 1, according to the proof in Proposition 20. The second item is not affected by the truncation so the bound is the same as for the truncated case, namely, greater or equal to $1-q(\frac{1}{2}\pm\frac{1}{2q})^M$. □

For the not truncated case, we have the following

Proposition 23. *With the same notations and assumptions as in Proposition 21,*

1. *If the samples are PLWE, Algorithm 9 guesses correctly with probability at least* p_0^M.
2. *If the samples are uniform, Algorithm 9 guesses correctly with probability at least* $1-q(\frac{1}{2}\pm\frac{1}{2q})^M$.

Proof. The first item is equivalent to computing $\mathrm{P}(E|D=\mathcal{G}_\sigma)$, which is at least p_0^M, according to the proof in Proposition 21. The second item is not affected by the truncation so the bound is the same as for the truncated case, namely, greater or equal to $1-q(\frac{1}{2}\pm\frac{1}{2q})^M$. □

An improvement for a higher number of samples. As discussed in Section 3.2, the extension algorithm presented which employed Algorithm 3 as a sub-process did not employ any feature specific of $\alpha\in\mathbb{F}_q$. As such, the same discussion and results presented there can be applied to finite extensions of $\mathbb{F}_q$, giving rise to the following algorithm:

Input:	A collection of M samples $S = \{(a_i(x), b_i(x))\}_{i=1}^{M} \in R_{q,0} \times R_q$
	A choice M_0 of the number of samples for the sub-process
Output:	A guess for the distribution of the samples, either
	PLWE or **UNIFORM**

- $T := \lceil \lfloor M/M_0 \rfloor \cdot p_0^{M_0} \rceil$
- $C := 0$
- *for* $j \in \{0, ..., \lfloor M/M_0 \rfloor - 1\}$ *do*
  - $result :=$ SmallValueAttackTraces $\left(S_j := \{(a_i(x), b_i(x))\}_{i=j\cdot M_0+1}^{(j+1)\cdot M_0}\right)$
    - *if* $result \neq$ **NOT PLWE** *then*
      - $C := C + 1$
- *if* $C < T$ *then return* **UNIFORM**
- *else return* **PLWE**

Algorithm 10. Algorithm for Extended Small Values Attack over finite extensions of $\mathbb{F}_q$

Moreover, since the success probabilities detailed in Proposition 23 for Algorithm 9 are the same as the ones in Proposition 9 for Algorithm 3, all the results detailed in Section 3.2 regarding the success probabilities of the extended attack apply here as well.

In particular, this means that:

- Case $D = \mathcal{G}_\sigma$: We have that

$$\mathrm{P}(C \geq T | D = \mathcal{G}_\sigma) := 1 - F(T-1, \lfloor M/M_0 \rfloor, \mathrm{P}_{\mathcal{G}_\sigma}(M_0)) \geq 1 - F(T-1, \lfloor M/M_0 \rfloor, p_0^{M_0})$$

- Case $D = U$: We have:

$$\mathrm{P}(C < T | D = U) := F(T-1, \lfloor M/M_0 \rfloor, \mathrm{P}_U(M_0)) \geq F(T-1, \lfloor M/M_0 \rfloor, 1 - (\frac{1}{2} \pm \frac{1}{2q})^{M_0})$$

Proposition 24. *Let C, T, M_0 be as specified in Algorithm 10. Then, if $1 - (\frac{1}{2} \pm \frac{1}{2q})^{M_0} < p_0^{M_0}$, Algorithm 10 is successful.*

Proof. See proof of Proposition 10. □

5.4 Unbounded Small Values Attack

This scenario represents the case in which the associated bound $2 \cdot \overline{\sigma} \leq \frac{q}{4}$ does not hold. Let E_i denote the event that $\mathrm{Tr}(b_i(\alpha) - g \cdot a_i(\alpha)) \in [\frac{-q}{4}, \frac{q}{4})$ for the i-th sample and fixed g. Then we must compute the probability of E_i given a PLWE sample distribution. But this can be done in the same way as in Section 3.3, just considering the Gaussian distributions derived from the application of the trace, as defined above.

Note that, as stated in Proposition 13, the trace operator respects the distributions. Accordingly, the application of the trace operator to the tentative error values in the cases the samples are either PLWE or uniform does not modify the expected behavior of the distribution they follow with respect to the original case of Section 3.

To integrate the trace evaluation into the attack, it is required to inspect the threshold value T, to verify that, under this new setting, it is still a useful decisional limit. Now denoting $g^* = s(\alpha) \in \mathbb{F}_{q^n}$, it is clear that for the any value $g \neq g^*$ we have

$$\begin{aligned}\frac{1}{n} \cdot \mathrm{Tr}(b(\alpha) - g \cdot a(\alpha)) &= \frac{1}{n} \cdot \mathrm{Tr}((s(\alpha) - g) \cdot a(\alpha) + e(\alpha)) \\ &= \frac{1}{n} \cdot \mathrm{Tr}((s(\alpha) - g) \cdot a(\alpha)) + \frac{1}{n} \cdot \mathrm{Tr}(e(\alpha)),\end{aligned}$$

which, by Proposition 13, is the sum of uniform value in $\mathbb{F}_q$, and a discrete Gaussian of mean 0 and variance $\overline{\sigma}^2$ in $\mathbb{F}_q$, respectively. Thus, we arrive to the same conditions as in Section 3.3.

Input: A collection of samples $S := \{(a_i(x), b_i(x))\}_{i=1}^{\ell} \subseteq R_{q,0} \times R_q$ according to a certain distribution.
A value δ representing the difference over $\frac{1}{2}$.
Output: A guess into the distribution of the samples, either **PLWE** or **UNIFORM**.

- $T := \left\lceil \frac{1}{2}\left(\ell \cdot q + 2\ell \cdot \delta \pm \ell \cdot \left(1 - \frac{1}{q}\right)\right)\right\rceil$
- $C := 0$
- *for* $g \in \mathbb{F}_q$ *do*
  - *for* $(a_i(x), b_i(x)) \in S$ *do*
    - *if* $\frac{1}{n}(\mathrm{Tr}(b_i(\alpha)) - a_i(\alpha)g) \in [-\frac{q}{4}, \frac{q}{4})$ *then*
      - $C = C + 1$
- *if* $C < T$ *then return* **UNIFORM**
- *else return* **PLWE**

Algorithm 11. PLWE Unbounded Small Values Attack

Denoting the probability as $\mathrm{P}(E_i|D = \mathcal{G}_\sigma) := \frac{1}{2} + \delta$, and referring to ℓ as the number of samples provided, the attack is described in Algorithm 11.

The probability of success remain the same as in Section 3.3. The only differences come from the fact that the Gaussian image generated will be smaller if we go to higher-degree extensions, though at the price of an increase in the associated computational cost.

Acknowledgments. I. Blanco-Chacón is partially supported by the Spanish National Research Plan, grant no.PID2022-136944NB-I00, by grant PID2019-104855RB-I00, funded by MCIN/AEI/10.13039/501100011033 and by the Universidad de Alcalá grant CCG20/IA-057. R. Durán-Díaz is partially supported by the Spanish National Research Plan, grant SATPQC (PID2024-156099OB-I00), funded by MCIN / AEI / 10.13039 / 501100011033. R. Martín Sánchez-Ledesma is partially supported by the PQReact Project. This project has received funding from the European Union's Horizon Europe research and innovation program under grant agreement no. 101119547.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

References

1. Barbero-Lucas, B., Blanco-Chacón, I., Durán-Díaz, R., Njah Nchiwo, R.Y.: Cryptanalysis of PLWE based on zero-trace quadratic roots. arXiv (2023). https://doi.org/10.48550/arXiv.2312.11533
2. Blanco-Chacón, I.: On the RLWE/PLWE equivalence for cyclotomic number fields. Applicable Algebra in Engineering, Communication and Computing (2020). https://doi.org/10.1007/s00200-020-00433-z
3. Blanco-Chacón, I., Durán-Díaz, R., Njah Nchiwo, R.Y., Barbero-Lucas, B.: Trace-based cryptanalysis of cyclotomic $R_{q,0} \times R_q$-PLWE for the non-split case. Communications in Mathematics **31**(2), 115–135 (Jul 2023). https://doi.org/10.46298/cm.11153
4. Blanco-Chacón, I., López-Hernanz, L.: RLWE/PLWE equivalence for the maximal totally real subextension of the $2^r pq$-th cyclotomic field. Advances in Mathematics of Communications pp. 1–21 (2022). https://doi.org/10.48550/arXiv.2111.13484
5. Blasco Lorenzo, A., Pérez Díaz, S.: Modelos aleatorios en Ingeniería. Ediciones Paraninfo (2015)
6. Castryck, W., Iliashenko, I., Vercauteren, F.: On Error Distributions in Ring-based LWE. Cryptology ePrint Archive, Report 2016/240 (2016), https://ia.cr/2016/240, https://ia.cr/2016/240
7. Castryck, W., Iliashenko, I., Vercauteren, F.: Provably Weak Instances of Ring-LWE Revisited. In: Fischlin, M., Coron, J. (eds.) Advances in Cryptology - EUROCRYPT 2016 - 35th Annual International Conference on the Theory and Applications of Cryptographic Techniques, Vienna, Austria, May 8-12, 2016, Proceedings, Part I. Lecture Notes in Computer Science, vol. 9665, pp. 147–167. Springer (2016). https://doi.org/10.1007/978-3-662-49890-3_6
8. Chen, H., Lauter, K., Stange, K.E.: Attacks on the Search RLWE Problem with Small Errors. SIAM Journal on Applied Algebra and Geometry **1**(1), 665–682 (2017). https://doi.org/10.1137/16M1096566

9. Cramer, R., Ducas, L., Wesolowski, B.: Short Stickelberger Class Relations and Application to Ideal-SVP. In: Coron, J., Nielsen, J.B. (eds.) Advances in Cryptology - EUROCRYPT 2017 - 36th Annual International Conference on the Theory and Applications of Cryptographic Techniques, Paris, France, April 30 - May 4, 2017, Proceedings, Part I. Lecture Notes in Computer Science, vol. 10210, pp. 324–348 (2017). https://doi.org/10.1007/978-3-319-56620-7_12
10. Di Scala, A.J., Sanna, C., Signorini, E.: On the condition number of the Vandermonde matrix of the nth cyclotomic polynomial. Journal of Mathematical Cryptology **15**(1), 174–178 (2021). https://doi.org/doi:10.1515/jmc-2020-0009
11. Ducas, L., Durmus, A.: Ring-LWE in Polynomial Rings. In: Fischlin, M., Buchmann, J., Manulis, M. (eds.) Public Key Cryptography – PKC 2012. pp. 34–51. Springer Berlin Heidelberg, Berlin, Heidelberg (2012). https://doi.org/10.1007/978-3-642-30057-8_3
12. Ducas, L., Plançon, M., Wesolowski, B.: On the Shortness of Vectors to Be Found by the Ideal-SVP Quantum Algorithm. In: Boldyreva, A., Micciancio, D. (eds.) Advances in Cryptology – CRYPTO 2019. vol. 11692, pp. 322–351. Springer International Publishing (2019). https://doi.org/10.1007/978-3-030-26948-7_12
13. Elias, Y., Lauter, K.E., Ozman, E., Stange, K.E.: Provably Weak Instances of Ring-LWE. In: Gennaro, R., Robshaw, M. (eds.) Advances in Cryptology – CRYPTO 2015. pp. 63–92. No. 9215 in Lecture Notes in Computer Science, Springer Berlin Heidelberg, Berlin, Heidelberg (2015). https://doi.org/10.1007/978-3-662-47989-6_4
14. Elias, Y., Lauter, K.E., Ozman, E., Stange, K.E.: Ring-LWE Cryptography for the Number Theorist. In: Eischen, E.E., Long, L., Pries, R., Stange, K.E. (eds.) Directions in Number Theory. Association for Women in Mathematics Series, vol. 3, pp. 271–290. Springer International Publishing, Cham (2016). https://doi.org/10.1007/978-3-319-30976-7_9
15. Lyubashevsky, V., Peikert, C., Regev, O.: On Ideal Lattices and Learning with Errors over Rings. Journal of the ACM **60**(6), 43:1–43:35 (November 2013). https://doi.org/10.1145/2535925
16. NIST: Module-Lattice-Based Digital Signature Standard. Tech. Rep. 204, National Institute of Standards and Technologies (2024). https://doi.org/10.6028/NIST.FIPS.204
17. NIST: Module-Lattice-Based Key-Encapsulation Mechanism Standard. Tech. Rep. 203, National Institute of Standards and Technologies (2024). https://doi.org/10.6028/NIST.FIPS.203
18. Peikert, C.: How (Not) to Instantiate Ring-LWE. In: Zikas, V., De Prisco, R. (eds.) Security and Cryptography for Networks. Lecture Notes in Computer Science, vol. 9841, pp. 411–430. Springer International Publishing (2016). https://doi.org/10.1007/978-3-319-44618-9_22
19. Rosca, M., Stehlé, D., Wallet, A.: On the Ring-LWE and Polynomial-LWE Problems. In: Nielsen, J.B., Rijmen, V. (eds.) Advances in Cryptology – EUROCRYPT 2018. pp. 146–173. Springer International Publishing, Cham (2018). https://doi.org/10.1007/978-3-319-78381-9_6
20. Stehlé, D., Steinfeld, R., Tanaka, K., Xagawa, K.: Efficient Public Key Encryption Based on Ideal Lattices. In: Matsui, M. (ed.) Advances in Cryptology – ASIACRYPT 2009. pp. 617–635. Springer Berlin Heidelberg, Berlin, Heidelberg (2009). https://doi.org/10.1007/978-3-642-10366-7_36
21. Stewart, I., Tall, D.: Algebraic Number Theory and Fermat's Last Theorem. A K Peters, Ltd., Natick, MA, USA, 3rd edn. (2002)

A Experimental results

We conducted also experiments to check the ability to find elements in $R_{q,0}$, fixing the following parameters:

– We choose $N = 64$ and the following irreducible polynomial in $\mathbb{Z}[x]$:

$$\begin{aligned} f(x) = x^{64} &+ 1384\,x^{63} + 3830\,x^{62} + 1279\,x^{61} + 3341\,x^{60} + 4034\,x^{59} + 3570\,x^{58} \\ &+ 1973\,x^{57} + 4111\,x^{56} + 337\,x^{55} + 3980\,x^{54} + 789\,x^{53} + 2903\,x^{52} + 2876\,x^{51} \\ &+ 644\,x^{50} + 4101\,x^{49} + 2944\,x^{48} + 959\,x^{47} + 1378\,x^{46} + 2161\,x^{45} + 3710\,x^{44} \\ &+ 3040\,x^{43} + 337\,x^{42} + 1329\,x^{41} + 3756\,x^{40} + 866\,x^{39} + 1790\,x^{38} + 3020\,x^{37} \\ &+ 2584\,x^{36} + 266\,x^{35} + 140\,x^{34} + 1952\,x^{33} + 1920\,x^{32} + 3463\,x^{31} + 2540\,x^{30} \\ &+ 1686\,x^{29} + 1825\,x^{28} + 1988\,x^{27} + 126\,x^{26} + 3366\,x^{25} + 2982\,x^{24} + 1958\,x^{23} \\ &+ 57\,x^{22} + 3543\,x^{21} + 1773\,x^{20} + 711\,x^{19} + 3607\,x^{18} + 2686\,x^{17} + 519\,x^{16} \\ &+ 1996\,x^{15} + 4106\,x^{14} + 3785\,x^{13} + 2775\,x^{12} + 1884\,x^{11} + 2988\,x^{10} + 1605\,x^{9} \\ &+ 1819\,x^{8} + 3000\,x^{7} + 3084\,x^{6} + 30\,x^{5} + 927\,x^{4} + 3110\,x^{3} + 2423\,x^{2} + 248\,x + 468 \end{aligned}$$

– $n = 4$, $q = 4133$, $a = 733$, so that the polynomial $g(x) = x^4 - 733$, is irreducible in $\mathbb{F}_q$, but $g(x)|f(x)$ in $\mathbb{F}_q[x]$. The order of a in $\mathbb{F}_q^*$ is $r = 4$.

Next we provide four elements in $R_{q,0}$, along with the number of invocations to a uniformly random oracle, needed to generate each of them:

1. Number of invocations: 25522319025. The element is:

$$\begin{aligned}&524x^{63} + 3572x^{62} + 1163x^{61} + 3674x^{60} + 1684x^{59} + 3017x^{58} + 2292x^{57} + 2054x^{56} +\\ &1126x^{55} + 143x^{54} + 2724x^{53} + 1763x^{52} + 3526x^{51} + 3950x^{50} + 75x^{49} + 1417x^{48} +\\ &296x^{47} + 977x^{46} + 3383x^{45} + 2113x^{44} + 3292x^{43} + 1557x^{42} + 2369x^{41} + 231x^{40} +\\ &609x^{39} + 2452x^{38} + 1369x^{37} + 2802x^{36} + 4032x^{35} + 2059x^{34} + 4021x^{33} + 2448x^{32} +\\ &342x^{31} + 1166x^{30} + 539x^{29} + 3599x^{28} + 2907x^{27} + 2083x^{26} + 1879x^{25} + 2551x^{24} +\\ &3151x^{23} + 2031x^{22} + 1529x^{21} + 304x^{20} + 555x^{19} + 2362x^{18} + 1182x^{17} + 1890x^{16} +\\ &3717x^{15} + 2592x^{14} + 1361x^{13} + 4123x^{12} + 51x^{11} + 3019x^{10} + 4082x^{9} + 1079x^{8} +\\ &1247x^{7} + 2353x^{6} + 2315x^{5} + 972x^{4} + 2298x^{3} + 3474x^{2} + 3553x + 1883\end{aligned}$$

2. Number of invocations: 79228885824. The element is:

$$\begin{aligned}&945x^{63} + 3353x^{62} + 1216x^{61} + 979x^{60} + 802x^{59} + 691x^{58} + 438x^{57} + 3922x^{56} +\\ &2532x^{55} + 3362x^{54} + 3378x^{53} + 1714x^{52} + 1625x^{51} + 2762x^{50} + 1708x^{49} + 2722x^{48} +\\ &3828x^{47} + 3220x^{46} + 3342x^{45} + 1421x^{44} + 1599x^{43} + 112x^{42} + 3893x^{41} + 2618x^{40} +\\ &3742x^{39} + 1342x^{38} + 37x^{37} + 995x^{36} + 260x^{35} + 2244x^{34} + 2638x^{33} + 141x^{32} +\\ &1145x^{31} + 1104x^{30} + 3886x^{29} + 3115x^{28} + 856x^{27} + 1159x^{26} + 2385x^{25} + 1265x^{24} +\\ &3609x^{23} + 3596x^{22} + 3689x^{21} + 2885x^{20} + 1665x^{19} + 411x^{18} + 2247x^{17} + 1700x^{16} +\\ &3362x^{15} + 2802x^{14} + 2342x^{13} + 1449x^{12} + 132x^{11} + 1247x^{10} + 3072x^{9} + 2475x^{8} +\\ &2811x^{7} + 2459x^{6} + 3916x^{5} + 991x^{4} + 1975x^{3} + 3335x^{2} + 1126x + 1083\end{aligned}$$

3. Number of invocations: 104055278839. The element is:

$$\begin{aligned}&4125x^{63} + 3199x^{62} + 1098x^{61} + 162x^{60} + 1695x^{59} + 1736x^{58} + 2288x^{57} + 3874x^{56} +\\ &1794x^{55} + 1281x^{54} + 1165x^{53} + 3769x^{52} + 1938x^{51} + 2190x^{50} + 1958x^{49} + 3578x^{48} +\\ &3234x^{47} + 382x^{46} + 201x^{45} + 3963x^{44} + 1463x^{43} + 1964x^{42} + 3641x^{41} + 3445x^{40} +\\ &3234x^{39} + 1866x^{38} + 3279x^{37} + 3788x^{36} + 2116x^{35} + 2851x^{34} + 3882x^{33} + 295x^{32} +\\ &1267x^{31} + 4009x^{30} + 1655x^{29} + 3892x^{28} + 1027x^{27} + 1389x^{26} + 1723x^{25} + 4019x^{24} +\\ &2587x^{23} + 3922x^{22} + 3290x^{21} + 1641x^{20} + 2646x^{19} + 251x^{18} + 642x^{17} + 4075x^{16} +\\ &27x^{15} + 50x^{14} + 1811x^{13} + 1647x^{12} + 64x^{11} + 3342x^{10} + 2699x^{9} + 2452x^{8} +\\ &3198x^{7} + 2745x^{6} + 2232x^{5} + 1859x^{4} + 1341x^{3} + 802x^{2} + 2162x + 838\end{aligned}$$

4. Number of invocations: 144485523325. The element is:

$$\begin{aligned}&1701x^{63} + 2817x^{62} + 3829x^{61} + 1416x^{60} + 1069x^{59} + 1079x^{58} + 1434x^{57} + 3596x^{56} +\\ &1019x^{55} + 1490x^{54} + 1603x^{53} + 4046x^{52} + 3686x^{51} + 3328x^{50} + 3621x^{49} + 1976x^{48} +\\ &3876x^{47} + 3670x^{46} + 1472x^{45} + 1442x^{44} + 1548x^{43} + 377x^{42} + 410x^{41} + 2707x^{40} +\\ &964x^{39} + 499x^{38} + 3991x^{37} + 1122x^{36} + 2234x^{35} + 1313x^{34} + 271x^{33} + 3444x^{32} +\\ &3496x^{31} + 2638x^{30} + 990x^{29} + 1222x^{28} + 488x^{27} + 2395x^{26} + 2825x^{25} + 2352x^{24} +\\ &2150x^{23} + 3640x^{22} + 2095x^{21} + 2489x^{20} + 3771x^{19} + 2760x^{18} + 439x^{17} + 2518x^{16} +\\ &2743x^{15} + 3259x^{14} + 3598x^{13} + 2738x^{12} + 1755x^{11} + 1917x^{10} + 3443x^{9} + 1570x^{8} +\\ &3431x^{7} + 3689x^{6} + 3078x^{5} + 2266x^{4} + 3869x^{3} + 1526x^{2} + 822x + 2986\end{aligned}$$

Remark that, according to Proposition 14 and subsequent comments, the expected number of invocations is $\mathcal{O}\left(N^{2(n-1)}\right)$, which for this case amounts to $64^6 \simeq 68719 \cdot 10^6$. The average value in our experiment yields roughly $88323 \cdot 10^6$ invocations, pretty near to the foreseen value. However completing the computation took from around four days (wall time) for the fastest sample to around three weeks (also wall time) for the slowest. Given these figures, it seems that an example with $n = 5$ is way beyond our possibilities.